

ALLAMA IQBAL OPEN UNIVERSITY, ISLAMABAD

Department of Mathematics ■ Faculty of Sciences

Business Mathematics

Solved Question Bank

Course Code **1429** (also issued as **5405**) ■ Units 1–9

BA ■ B.Com ■ ADC ■ BS

What this document is. Full worked solutions to the SELF ASSESSMENT QUESTIONS printed at the end of each of the nine units of the official AIOU course book *Business Mathematics*, Course Code 5405/1429, Department of Mathematics, AIOU Islamabad (2023 printing).

Why these questions. AIOU sets its semester assignments and its final paper from this pool. Solving the book's own end-of-unit questions is therefore not a proxy for past-paper practice — it *is* past-paper practice, one step upstream of the paper.

Status. Prepared for private study. It is not an AIOU publication, carries no official endorsement, and must not be copied into an assignment for submission — see *How to use this* on page ii.

Provenance and honest limits

Where the questions came from

Every question in this document is reproduced from the end-of-unit SELF ASSESSMENT QUESTIONS of the official AIOU course book:

Business Mathematics, Course Code 5405/1429, Units 1–9.
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2023 printing. Distributed by AIOU at online.aiou.edu.pk.

I went looking for scanned past papers first. Copies of the Spring 2019, Spring 2020, Spring 2022 and Autumn 2023 papers for 1429 are circulating on several study sites, but every one of them is published as a photograph or a locked PDF with no recoverable text, so none could be transcribed accurately. The course book, by contrast, is published by AIOU itself with a real text layer, and its end-of-unit sets are the pool the assignments and the paper are drawn from. That is the source used here, and it is a better one.

Two places where the printed book is defective

The 2023 printing has typesetting faults, and this document does not paper over them:

- **Unit 6, Question 5** (the electronics materials problem) prints identical material requirements for resistors and transistors. That makes the coefficient matrix singular and the system inconsistent, so the question as printed has no solution. Rather than quietly patch the numbers, Section 6.5 shows the determinant vanishing, explains what a zero determinant means for a production plan, and then solves the repaired problem.
- **Unit 8, Question 7** (the production function) prints $0.8l^3$, which makes P unbounded above, so no maximising l exists. Section 8.6 works the second-order test on the printed function, states plainly that there is no maximum, and then solves the intended $0.8l^2 - 0.02l^3$ form.

A handful of sub-parts in Units 7 and 9 are illegible in the distributed PDF (radicals and fraction bars collapsed in the text layer). Those are named where they occur and skipped rather than guessed at.

How to use this

Read the question. Cover the solution. Work it on paper. *Then* compare. Reading a worked solution feels like learning and is not; the transfer to an exam hall comes almost entirely from the failed attempt that precedes it.

Three features are built for that loop:

Answer blocks The final result is boxed in green, so you can check whether you got there without reading how.

Check lines Most solutions substitute the answer back into the original equation. AIOU awards method marks; a substitution check is the cheapest way to catch a sign error before the marker does.

Where marks are lost The specific, recurring mistake for that question type, in red. These are worth more than the solutions.

On submitting this as an assignment

Do not. AIOU runs plagiarism checks and these solutions are, by design, in circulation. The purpose here is to let you check your own working and find the step where your method breaks. Copy the method, submit your own arithmetic.

What the nine units are worth

The paper draws across all nine units, but not evenly. The pattern below is what the book's own weighting and the circulating past papers agree on.

Unit	Topic	Typical exam load	The one thing it tests
1	Probability theory	Heavy	Addition vs. multiplication rule
2	Random variables	Medium	Probabilities must sum to 1
3	Equations and inequalities	Medium	Sign flips; two cases for $ \cdot $
4	Linear equations	Heavy	Slope, then a point
5	Matrices	Heavy	Conformability of the product
6	Determinants and inverses	Heavy	Cofactor signs; Cramer's rule
7	Differentiation	Heavy	Chain rule inside quotient rule
8	Partial derivatives	Medium	Hold the other variable constant
9	Optimization	Heavy	Second-order test, always

Units 4, 5, 6, 7 and 9 between them carry most of the marks and all of the long questions. If revision time is short, that is the order to spend it in.

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Probability Theory

Source: *Business Mathematics* 5405/1429, §1.9 SELF ASSESSMENT QUESTIONS, pp. 25–26.

The four rules the whole unit runs on

Addition (general) :	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Addition (exclusive) :	$P(A \cup B) = P(A) + P(B)$ when $A \cap B = \emptyset$
Multiplication (indep.) :	$P(A \cap B) = P(A) P(B)$
Conditional :	$P(A B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$

Everything below is one of these four, or a complement $P(A') = 1 - P(A)$.

1.1 Sample space of an odd draw

Q.1

Find the sample space for choosing an odd number from 1 to 15 at random.

Solution

The sample space is the list of every outcome the experiment can produce. Here the experiment is “pick one odd number between 1 and 15”, so the outcomes are the odd integers in that range:

$$S = \{1, 3, 5, 7, 9, 11, 13, 15\}, \quad n(S) = 8.$$

Each is equally likely, so any single outcome has probability $1/8$.

Answer. $S = \{1, 3, 5, 7, 9, 11, 13, 15\}$, with $n(S) = 8$.

1.2 Mutually exclusive versus collectively exhaustive

Q.2

What is the difference between mutually exclusive events and collectively exhaustive events?

Solution

These describe two different things: one is about *overlap*, the other about *coverage*. They are independent properties, and a set of events can have either, both, or neither.

	Mutually exclusive	Collectively exhaustive
Condition	$A \cap B = \emptyset$	$A \cup B = S$
In probability	$P(A \cap B) = 0$	$P(A \cup B) = 1$
Plain reading	They cannot both happen.	At least one must happen.
Concerns	Overlap between events	Coverage of the sample space

Take a single roll of a die, $S = \{1, 2, 3, 4, 5, 6\}$:

- $A = \{\text{even}\}$, $B = \{\text{odd}\}$ — both mutually exclusive *and* collectively exhaustive. No overlap, and together they cover S .
- $A = \{1, 2, 3\}$, $B = \{3, 4, 5, 6\}$ — collectively exhaustive but *not* mutually exclusive: the outcome 3 is in both.
- $A = \{1\}$, $B = \{2\}$ — mutually exclusive but *not* collectively exhaustive: a roll of 5 is in neither.

Answer. Mutually exclusive means the events cannot occur together ($A \cap B = \emptyset$). Collectively exhaustive means one of them must occur ($A \cup B = S$). The first is a statement about overlap, the second about coverage; neither implies the other.

Where marks are lost

The simplified addition rule $P(A \cup B) = P(A) + P(B)$ is valid *only* for mutually exclusive events. Applying it to overlapping events double-counts the intersection and is the single most common error in this unit.

1.3 Three independent admissions

Q.3

The probability that an applicant for pilot school will be admitted is 0.5. If three applicants are selected at random, what is the probability that (a) all three will be admitted, (b) none will be admitted, (c) only one will be admitted?

Solution

The three applicants are selected at random and independently, so $P(\text{admit}) = 0.5$ and $P(\text{reject}) = 0.5$ for each, and probabilities multiply across applicants.

(a) **All three admitted.** One way this can happen, and the three events are independent:

$$P(3 \text{ admitted}) = (0.5)(0.5)(0.5) = (0.5)^3 = 0.125.$$

(b) **None admitted.** Again one arrangement:

$$P(0 \text{ admitted}) = (0.5)^3 = 0.125.$$

(c) **Exactly one admitted.** Now the count matters. The admitted applicant can be the first, the second or the third — three arrangements, each of probability $(0.5)(0.5)(0.5) = 0.125$:

$$P(1 \text{ admitted}) = \binom{3}{1}(0.5)^1(0.5)^2 = 3(0.125) = 0.375.$$

✓ *Check:* The four cases 0, 1, 2, 3 admitted must exhaust the sample space: $0.125 + 0.375 + 0.375 + 0.125 = 1$. They do.

Answer. (a) 0.125 (b) 0.125 (c) 0.375

Where marks are lost

Part (c) is where marks go. “Only one” is not $(0.5)(0.5)^2$ — you must multiply by $\binom{3}{1} = 3$ to count *which* applicant it was. Parts (a) and (b) need no such factor because there is only one way for all or none.

1.4 Grades: complements of exclusive events

Q.4

A student estimates the probability of attaining an A in a maths course at 0.4 and the probability of attaining a B at 0.3. What is the probability that he/she (a) will not receive an A, (b) will not receive a B, (c) will receive neither an A nor a B?

Solution

Let A be the event of an A grade and B of a B grade, with $P(A) = 0.4$ and $P(B) = 0.3$. A single course yields a single grade, so A and B are mutually exclusive: $P(A \cap B) = 0$.

(a) By the complement rule, $P(A') = 1 - P(A) = 1 - 0.4 = 0.6$.

(b) $P(B') = 1 - P(B) = 1 - 0.3 = 0.7$.

(c) “Neither A nor B” is the complement of “A or B”. Because the two are mutually exclusive the addition rule loses its correction term:

$$P(A \cup B) = P(A) + P(B) = 0.4 + 0.3 = 0.7,$$

$$P((A \cup B)') = 1 - 0.7 = 0.3.$$

Answer. (a) 0.6 (b) 0.7 (c) 0.3

Where marks are lost

Part (c) is *not* $P(A') \times P(B') = 0.6 \times 0.7 = 0.42$. That multiplication assumes A' and B' are independent, and they are not — one grade excludes the other. Use the complement of the union.

1.5 Urn of dotted and striped balls

Q.5

An urn contains 8 green dotted balls, 10 green striped balls, 12 blue dotted balls and 10 blue striped balls. If a ball is selected at random from the urn, what is the probability that the ball will be (a) green or striped, (b) dotted, (c) blue or dotted?

Solution

Lay the counts out as a two-way table first. Almost every mark in this question is won or lost at this step.

	Dotted	Striped	Total
Green	8	10	18
Blue	12	10	22
Total	20	20	40

(a) **Green or striped.** These overlap — the 10 green striped balls are in both — so the general addition rule applies:

$$P(G \cup S) = P(G) + P(S) - P(G \cap S) = \frac{18}{40} + \frac{20}{40} - \frac{10}{40} = \frac{28}{40} = 0.70.$$

(b) **Dotted.** Straight off the column total:

$$P(D) = \frac{20}{40} = 0.50.$$

(c) **Blue or dotted.** Overlap is the 12 blue dotted balls:

$$P(B \cup D) = \frac{22}{40} + \frac{20}{40} - \frac{12}{40} = \frac{30}{40} = 0.75.$$

✓ *Check:* Counting directly: green or striped = $8+10+10 = 28$ of 40, and blue or dotted = $12+10+8 = 30$ of 40. Both agree.

Answer. (a) $28/40 = 0.70$ (b) $20/40 = 0.50$ (c) $30/40 = 0.75$

1.6 Four consecutive sixes

Q.6

A single die is rolled and each side has an equal chance of occurring. What is the probability of rolling four consecutive 6s?

Solution

Each roll is independent of the others — a die has no memory — and $P(6) = \frac{1}{6}$ on every roll. Independence lets the probabilities multiply:

$$P(\text{four 6s}) = \left(\frac{1}{6}\right)^4 = \frac{1}{1296} \approx 0.00077.$$

Answer. $\frac{1}{1296} \approx 0.00077$, i.e. about 8 times in every 10,000 attempts.

1.7 Conditional probability in a mixed urn

Q.7

A ball is selected at random from an urn containing three red striped balls, eight solid [red] balls, six yellow striped balls, four solid yellow balls and four blue striped balls. (a) What is the probability that the ball is yellow, given that it is striped? (b) What is the probability that the ball is striped, given that it is red? (c) What is the probability that the ball is blue, given that it is solid coloured?

The book prints “eight solid balls” without a colour. The colour is red: it is the only one of the three that would otherwise have no solid entry, and parts (b) and (c) both depend on it. That reading is used here and is flagged in the working.

Solution

Tabulate, with a zero where the book lists nothing:

	Striped	Solid	Total
Red	3	8	11
Yellow	6	4	10
Blue	4	0	4
Total	13	12	25

A conditional probability restricts attention to one row or one column, and renormalises within it.

(a) $P(\text{yellow} \mid \text{striped})$. The condition confines us to the striped column, which holds 13 balls; 6 of them are yellow:

$$P(Y \mid S) = \frac{P(Y \cap S)}{P(S)} = \frac{6/25}{13/25} = \frac{6}{13} \approx 0.4615.$$

(b) $P(\text{striped} \mid \text{red})$. Confine to the red row, 11 balls, of which 3 are striped:

$$P(S \mid R) = \frac{3/25}{11/25} = \frac{3}{11} \approx 0.2727.$$

(c) $P(\text{blue} \mid \text{solid})$. Confine to the solid column, 12 balls. None is blue:

$$P(B \mid \text{solid}) = \frac{0/25}{12/25} = 0.$$

This is a genuine answer, not a failure of the method: conditioning on “solid” makes “blue” impossible, because every blue ball in the urn is striped.

Answer. (a) $6/13 \approx 0.462$ (b) $3/11 \approx 0.273$ (c) 0

Where marks are lost

The denominator of a conditional probability is the *condition*, never the grand total. Writing $P(Y \mid S) = 6/25$ instead of $6/13$ throws away the entire point of the question.

1.8 Recovering an intersection from a union

Q.8

Suppose that E and F are events with $P(E) = 0.2$, $P(F) = 0.5$ and $P(E \cup F) = 0.6$. Determine (a) $P(E \cap F)$, (b) $P(E | F)$, (c) $P(F | E)$.

Solution

(a) Rearrange the general addition rule to make the intersection the subject:

$$P(E \cup F) = P(E) + P(F) - P(E \cap F) \implies P(E \cap F) = P(E) + P(F) - P(E \cup F),$$

$$P(E \cap F) = 0.2 + 0.5 - 0.6 = 0.1.$$

(b)

$$P(E | F) = \frac{P(E \cap F)}{P(F)} = \frac{0.1}{0.5} = 0.2.$$

(c)

$$P(F | E) = \frac{P(E \cap F)}{P(E)} = \frac{0.1}{0.2} = 0.5.$$

A remark worth a mark. Notice that $P(E | F) = 0.2 = P(E)$ and $P(F | E) = 0.5 = P(F)$: knowing one event tells you nothing about the other. Equivalently $P(E)P(F) = (0.2)(0.5) = 0.1 = P(E \cap F)$. So E and F are *independent*. If a follow-up part asks you to test independence, this is the test.

Answer. (a) 0.1 (b) 0.2 (c) 0.5; and E, F are independent, since $P(E \cap F) = P(E)P(F)$.

1.9 Three kings without replacement

Q.9

What is the probability of drawing three cards, without replacement, from a deck of cards and getting three kings?

Solution

Without replacement means the deck shrinks and the count of kings shrinks with it, so each factor must be recomputed:

Draw	Kings left	Cards left	Probability
1st	4	52	4/52
2nd	3	51	3/51
3rd	2	50	2/50

By the multiplication rule for dependent events,

$$P(3 \text{ kings}) = \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{2}{50} = \frac{24}{132600} = \frac{1}{5525} \approx 0.000181.$$

✓ *Check:* By combinations, $\binom{4}{3} / \binom{52}{3} = 4/22100 = 1/5525$. Same.

Answer. $\frac{1}{5525} \approx 0.00018$

Where marks are lost

Using $\left(\frac{4}{52}\right)^3$ treats the draws as *with* replacement and gives 0.000455 — more than twice the true value. Read the words “without replacement” as an instruction to change the denominator every line.

1.10 Conditioning on stripes

Q.10

An urn contains 18 red balls, 14 red striped balls, 16 yellow balls and 12 yellow striped balls. (a) Given that a ball selected from the urn is striped, what is the probability it is yellow? (b) Given that a ball selected from the urn is not striped, what is the probability it is red?

Solution

“18 red balls” means plain red, as distinct from the 14 red striped:

	Striped	Not striped	Total
Red	14	18	32
Yellow	12	16	28
Total	26	34	60

(a) Condition on the striped column (26 balls), of which 12 are yellow:

$$P(Y | S) = \frac{12/60}{26/60} = \frac{12}{26} = \frac{6}{13} \approx 0.4615.$$

(b) Condition on the not-striped column (34 balls), of which 18 are red:

$$P(R | S') = \frac{18/60}{34/60} = \frac{18}{34} = \frac{9}{17} \approx 0.5294.$$

Answer. (a) $6/13 \approx 0.462$ (b) $9/17 \approx 0.529$

Random Variables

Source: *Business Mathematics* 5405/1429, §2.7 SELF ASSESSMENT QUESTIONS, pp. 42–43.

What makes a list of numbers a probability distribution

A function $P(X = x)$ is a discrete probability distribution if and only if

$$0 \leq P(X = x) \leq 1 \quad \text{for every } x, \quad \text{and} \quad \sum_{\text{all } x} P(X = x) = 1.$$

The second condition is the whole of Questions 8, 9 and 10, and it is the check that catches an arithmetic slip in Questions 3 to 7.

2.1 Defining a random variable

Q.1

Define random variable. What is the difference between a discrete random variable and a continuous random variable?

Solution

A **random variable** is a rule that assigns one real number to each outcome of a random experiment. Formally it is a function $X : S \rightarrow \mathbb{R}$, where S is the sample space. It converts outcomes that may not be numeric (“head”, “defective”) into numbers that can be averaged, added and distributed.

	Discrete	Continuous
Values taken	A countable set — typically integers	Every value in an interval
Arises from	Counting	Measuring
Probability of a point	$P(X = x)$ can be positive	$P(X = x) = 0$ always
Described by	Probability mass function	Probability density function $f(x)$
Total probability	$\sum_x P(X = x) = 1$	$\int_{-\infty}^{\infty} f(x) dx = 1$
Business example	Number of defective units in a batch	Time to serve a customer

Answer. A random variable assigns a real number to each outcome of an experiment. It is discrete if its values are countable (results of counting) and continuous if it can take any value in an interval (results of measuring). For a continuous variable $P(X = x) = 0$ for every single point, so probability is found over intervals via a density function.

2.2 Discrete random variables in business

Q.2

Write down the applications of the discrete random variable in business mathematics.

Solution

Any business quantity that is obtained by *counting* is a discrete random variable, and each supports a standard decision:

Quality control The number of defective items in a sample of n is binomial. It drives accept/reject rules for incoming batches.

Demand and inventory Units of a product sold per day is discrete, and its distribution sets the re-order level and the safety stock.

Queue arrivals Customers arriving at a counter per hour is typically Poisson, which fixes how many tellers to staff.

Insurance and warranty The number of claims in a period is discrete; its expected value is the premium loading.

Credit and default The number of accounts defaulting out of n loans is binomial, and drives the provision held against the portfolio.

Expected monetary value For any of the above, $E(X) = \sum x P(X = x)$ turns the distribution into the single number a manager decides on.

Answer. Quality control (defectives per batch), demand and inventory planning, arrivals in queueing and staffing, insurance claim counts, and default counts in credit risk — each summarised for decision-making by $E(X) = \sum x P(X = x)$.

2.3 False alarms: frequency to probability

Q.3

The fire chief for a small volunteer fire department has compiled data on the number of false alarms called in each day for the past 360 days. The frequency distribution below summarises the findings. Construct the probability distribution for this study.

No. of false alarms	Frequency
0	75
1	80
2	77
3	40
4	28
5	24
6	20
7	16
Total	360

Solution

Converting a frequency distribution into a probability distribution is one division, applied to every row: relative frequency estimates probability,

$$P(X = x) = \frac{f(x)}{\sum f} = \frac{f(x)}{360}.$$

x	$f(x)$	$P(X = x) = f/360$	Decimal
0	75	75/360	0.2083
1	80	80/360	0.2222
2	77	77/360	0.2139
3	40	40/360	0.1111
4	28	28/360	0.0778
5	24	24/360	0.0667
6	20	20/360	0.0556
7	16	16/360	0.0444
Total	360	1	1.0000

✓ *Check:* $75 + 80 + 77 + 40 + 28 + 24 + 20 + 16 = 360$, so the probabilities sum to $360/360 = 1$ exactly. The distribution is valid.

As a by-product, the expected number of false alarms per day is

$$\begin{aligned} E(X) &= \sum x P(X = x) = \frac{0(75) + 1(80) + 2(77) + 3(40) + 4(28) + 5(24) + 6(20) + 7(16)}{360} \\ &= \frac{818}{360} \approx 2.27 \text{ false alarms per day.} \end{aligned}$$

Answer. $P(X = x) = f(x)/360$ for $x = 0, \dots, 7$, giving 0.2083, 0.2222, 0.2139, 0.1111, 0.0778, 0.0667, 0.0556, 0.0444, which sum to 1. Mean ≈ 2.27 false alarms per day.

2.4 Three tosses of a fair coin

Q.4

Construct the discrete probability distribution which corresponds to the experiment of tossing a fair coin three times. Let X equal the number of heads occurring in three tosses. What is the probability of two or more heads?

Solution

There are $2^3 = 8$ equally likely outcomes. Listing them makes the counts undeniable:

x (heads)	Outcomes	Count	$P(X = x)$
0	TTT	1	1/8
1	HTT, THT, TTH	3	3/8
2	HHT, HTH, THH	3	3/8
3	HHH	1	1/8
Total		8	1

Equivalently, $X \sim \text{Binomial}(n = 3, p = \frac{1}{2})$ and $P(X = x) = \binom{3}{x} \left(\frac{1}{2}\right)^3$.

Two or more heads means $X = 2$ or $X = 3$, which are mutually exclusive:

$$P(X \geq 2) = P(2) + P(3) = \frac{3}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2} = 0.5.$$

Answer. $P(0) = 1/8$, $P(1) = 3/8$, $P(2) = 3/8$, $P(3) = 1/8$; and $P(X \geq 2) = 1/2 = 0.5$.

2.5 Four balls drawn without replacement

Q.5

A bag contains 4 red and 6 black balls. A sample of 4 balls is selected from the bag without replacement. Let X be the number of red balls. Find the probability distribution of X .

Solution

Sampling *without* replacement from a finite bag makes this hypergeometric, not binomial. With 4 red and 6 black in a bag of 10, drawing 4:

$$P(X = x) = \frac{\binom{4}{x} \binom{6}{4-x}}{\binom{10}{4}}, \quad x = 0, 1, 2, 3, 4.$$

The denominator is $\binom{10}{4} = 210$ for every row.

x	Numerator	Value	$P(X = x)$	Decimal
0	$\binom{4}{0} \binom{6}{4} = 1 \times 15$	15	$15/210 = 1/14$	0.0714
1	$\binom{4}{1} \binom{6}{3} = 4 \times 20$	80	$80/210 = 8/21$	0.3810
2	$\binom{4}{2} \binom{6}{2} = 6 \times 15$	90	$90/210 = 3/7$	0.4286
3	$\binom{4}{3} \binom{6}{1} = 4 \times 6$	24	$24/210 = 4/35$	0.1143
4	$\binom{4}{4} \binom{6}{0} = 1 \times 1$	1	$1/210$	0.0048
Total		210	1	1.0000

✓ *Check:* $15 + 80 + 90 + 24 + 1 = 210 = \binom{10}{4}$, so the probabilities sum to 1.

Answer. $P(0) = \frac{1}{14}$, $P(1) = \frac{8}{21}$, $P(2) = \frac{3}{7}$, $P(3) = \frac{4}{35}$, $P(4) = \frac{1}{210}$.

Where marks are lost

Treating this as binomial with $p = 4/10$ gives $P(2) = 0.3456$ instead of 0.4286. Binomial requires a constant p , which only holds if the ball is replaced. “Without replacement” means hypergeometric.

2.6 Three balls from five white and three black

Q.6

Three balls are drawn from a bag containing 5 white and 3 black balls. If X represents the number of white balls drawn, find the probability distribution of X .

Solution

Same structure as Q.5, with $\binom{8}{3} = 56$ as the denominator:

$$P(X = x) = \frac{\binom{5}{x} \binom{3}{3-x}}{\binom{8}{3}}, \quad x = 0, 1, 2, 3.$$

x	Numerator	Value	$P(X = x)$	Decimal
0	$\binom{5}{0} \binom{3}{3} = 1 \times 1$	1	$1/56$	0.0179
1	$\binom{5}{1} \binom{3}{2} = 5 \times 3$	15	$15/56$	0.2679
2	$\binom{5}{2} \binom{3}{1} = 10 \times 3$	30	$30/56 = 15/28$	0.5357
3	$\binom{5}{3} \binom{3}{0} = 10 \times 1$	10	$10/56 = 5/28$	0.1786
Total		56	1	1.0000

✓ Check: $1 + 15 + 30 + 10 = 56 = \binom{8}{3}$.

Answer. $P(0) = \frac{1}{56}$, $P(1) = \frac{15}{56}$, $P(2) = \frac{15}{28}$, $P(3) = \frac{5}{28}$.

2.7 Five rolls of a die

Q.7

A fair die is rolled 5 times. Let X represent the number of times the face 3 turns up. Obtain the probability distribution of X .

Solution

Here the rolls *are* independent and $p = \frac{1}{6}$ does not change, so this is binomial — the contrast with Questions 5 and 6 is deliberate:

$$X \sim \text{Binomial}\left(n = 5, p = \frac{1}{6}\right), \quad P(X = x) = \binom{5}{x} \left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{5-x}.$$

Every term has denominator $6^5 = 7776$.

x	Working	$P(X = x)$	Decimal
0	$1 \cdot 5^5$	$3125/7776$	0.4019
1	$5 \cdot 5^4$	$3125/7776$	0.4019
2	$10 \cdot 5^3$	$1250/7776$	0.1608
3	$10 \cdot 5^2$	$250/7776$	0.0322
4	$5 \cdot 5^1$	$25/7776$	0.0032
5	$1 \cdot 5^0$	$1/7776$	0.0001
Total		7776/7776	1.0000

✓ Check: $3125 + 3125 + 1250 + 250 + 25 + 1 = 7776 = 6^5$.

Note $E(X) = np = 5 \times \frac{1}{6} = \frac{5}{6} \approx 0.83$, consistent with the mass piling up at $x = 0$ and $x = 1$.

Answer. $P(X = x) = \binom{5}{x}(1/6)^x(5/6)^{5-x}$, giving 0.4019, 0.4019, 0.1608, 0.0322, 0.0032, 0.0001 for $x = 0, \dots, 5$.

2.8 Three missing-value problems**Q.8, Q.9, Q.10**

Q.8 The distribution of X over 1, 2, 3, 4, 5 is 0.1, 0.3, y , 0.2, 0.1. Find y .

Q.9 Find k such that the values 0.01, 0.25, 0.40, k , 0.04 over $X = 2, 3, 4, 5, 6$ form a probability distribution.

Q.10 For $X = 0, 2, 4, 6, 8$ the probabilities are $K, 2K, 4K, 2K, K$. Rewrite the distribution after finding K .

Solution

All three are the same one-line idea: the probabilities must total 1.

Q.8.

$$0.1 + 0.3 + y + 0.2 + 0.1 = 1 \implies 0.7 + y = 1 \implies y = 0.3.$$

Q.9.

$$0.01 + 0.25 + 0.40 + k + 0.04 = 1 \implies 0.70 + k = 1 \implies k = 0.30.$$

Q.10.

$$K + 2K + 4K + 2K + K = 1 \implies 10K = 1 \implies K = 0.1.$$

The completed distribution is symmetric about $x = 4$:

x	0	2	4	6	8
$P(X = x)$	0.1	0.2	0.4	0.2	0.1

✓ *Check:* $0.1 + 0.2 + 0.4 + 0.2 + 0.1 = 1.0$; every value lies in $[0, 1]$. Both conditions hold. By symmetry $E(X) = 4$.

Answer. Q.8: $y = 0.3$. Q.9: $k = 0.30$. Q.10: $K = 0.1$, giving 0.1, 0.2, 0.4, 0.2, 0.1.

Where marks are lost

After solving for the unknown, confirm it lies in $[0, 1]$. A negative value means the other probabilities were misread — the answer is not “ $k = -0.2$ ”, it is “no such k exists”.

Equations and Inequalities

Source: *Business Mathematics* 5405/1429, §3.9 SELF ASSESSMENT QUESTIONS, pp. 64–65.

Three tools for the whole unit

Quadratic formula : $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ for $ax^2 + bx + c = 0$

Absolute value : $|u| = |v| \iff u = v \text{ or } u = -v$

Distance / midpoint : $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

And one rule that costs more marks than any formula: **multiplying or dividing an inequality by a negative number reverses the inequality sign.**

3.1 First-degree equations

Q.1

Solve the following first-degree equations. (a) $8x - 6 = 5x + 3$ (b) $-15 + 35x = 8x - 9$
(c) $(x + 9) - (-6 + 4x) + 4 = 0$ (d) $\frac{y}{8} - 10 = \frac{y}{4} - 9$

Solution

(a) Collect x on the left, constants on the right:

$$8x - 5x = 3 + 6 \implies 3x = 9 \implies x = 3.$$

✓ Check: $8(3) - 6 = 18$ and $5(3) + 3 = 18$.

(b)

$$35x - 8x = -9 + 15 \implies 27x = 6 \implies x = \frac{6}{27} = \frac{2}{9}.$$

✓ Check: LHS = $-15 + 35(\frac{2}{9}) = -15 + \frac{70}{9} = -\frac{65}{9}$; RHS = $8(\frac{2}{9}) - 9 = \frac{16}{9} - 9 = -\frac{65}{9}$.

(c) Clear the brackets, watching the sign on the second one:

$$(x + 9) - (-6 + 4x) + 4 = x + 9 + 6 - 4x + 4 = -3x + 19.$$

$$-3x + 19 = 0 \implies x = \frac{19}{3}.$$

(d) Multiply through by the LCD, 8, before doing anything else:

$$y - 80 = 2y - 72 \implies -80 + 72 = 2y - y \implies y = -8.$$

✓ Check: LHS = $\frac{-8}{8} - 10 = -11$; RHS = $\frac{-8}{4} - 9 = -11$.

Answer. (a) $x = 3$ (b) $x = \frac{2}{9}$ (c) $x = \frac{19}{3}$ (d) $y = -8$

Where marks are lost

Part (c): $-(-6 + 4x)$ is $+6 - 4x$, not $-6 - 4x$. The minus sign distributes over *both* terms in the bracket.

3.2 Quadratics by factorisation

Q.2

Solve the following quadratic equations using the factorisation method. (a) $t^2 + 4t = 21$ (b) $4z^2 + 18z - 10 = 0$

Solution

(a) Move everything to one side first — factorisation only works against zero:

$$t^2 + 4t - 21 = 0.$$

Two numbers multiplying to -21 and adding to $+4$: these are $+7$ and -3 .

$$(t + 7)(t - 3) = 0 \implies t = -7 \text{ or } t = 3.$$

✓ Check: $(-7)^2 + 4(-7) = 49 - 28 = 21$; $3^2 + 4(3) = 9 + 12 = 21$.

(b) Take out the common factor 2 to keep the numbers small:

$$4z^2 + 18z - 10 = 0 \implies 2(2z^2 + 9z - 5) = 0 \implies 2z^2 + 9z - 5 = 0.$$

Split the middle term: $2 \times (-5) = -10$, and $+10$ and -1 give $+9$:

$$2z^2 + 10z - z - 5 = 0 \implies 2z(z + 5) - 1(z + 5) = 0 \implies (2z - 1)(z + 5) = 0.$$

$$z = \frac{1}{2} \text{ or } z = -5.$$

✓ Check: $4(\frac{1}{2}) + 18(\frac{1}{2}) - 10 = 1 + 9 - 10 = 0$; $4(25) + 18(-5) - 10 = 100 - 90 - 10 = 0$.

Answer. (a) $t = -7, 3$ (b) $z = \frac{1}{2}, -5$

3.3 Quadratics by formula

Q.3

Solve using the quadratic formula. (a) $4x^2 + 3x - 1 = 0$ (b) $4t^2 - 64 = 0$

Solution

(a) $a = 4$, $b = 3$, $c = -1$. Compute the discriminant first — it tells you what kind of answer to expect:

$$b^2 - 4ac = 3^2 - 4(4)(-1) = 9 + 16 = 25 > 0 \quad (\text{two distinct real roots}).$$

$$x = \frac{-3 \pm \sqrt{25}}{2(4)} = \frac{-3 \pm 5}{8} \implies x = \frac{2}{8} = \frac{1}{4} \text{ or } x = \frac{-8}{8} = -1.$$

✓ *Check:* $4(\frac{1}{16}) + 3(\frac{1}{4}) - 1 = \frac{1}{4} + \frac{3}{4} - 1 = 0$; $4(1) + 3(-1) - 1 = 0$.

(b) $a = 4$, $b = 0$, $c = -64$:

$$t = \frac{0 \pm \sqrt{0 - 4(4)(-64)}}{8} = \frac{\pm \sqrt{1024}}{8} = \frac{\pm 32}{8} = \pm 4.$$

Faster without the formula: $4t^2 = 64 \Rightarrow t^2 = 16 \Rightarrow t = \pm 4$.

Answer. (a) $x = \frac{1}{4}, -1$ (b) $t = \pm 4$

Where marks are lost

In (b) the answer is $t = \pm 4$, not $t = 4$. Taking a square root of both sides of $t^2 = 16$ produces two roots; dropping the negative one loses half the marks.

3.4 Completing the square

Q.4

Solve using the completing-the-square method. (a) $2y^2 + 5y - 3 = 0$ (b) $s^2 + s - 3 = 0$

Solution

(a) Divide through by the leading coefficient so that y^2 stands alone, then move the constant across:

$$y^2 + \frac{5}{2}y - \frac{3}{2} = 0 \Rightarrow y^2 + \frac{5}{2}y = \frac{3}{2}.$$

Half the coefficient of y is $\frac{5}{4}$; add its square $\frac{25}{16}$ to both sides:

$$y^2 + \frac{5}{2}y + \frac{25}{16} = \frac{3}{2} + \frac{25}{16} = \frac{24}{16} + \frac{25}{16} = \frac{49}{16},$$

$$\left(y + \frac{5}{4}\right)^2 = \frac{49}{16} \Rightarrow y + \frac{5}{4} = \pm \frac{7}{4} \Rightarrow y = \frac{-5 \pm 7}{4}.$$

$$y = \frac{2}{4} = \frac{1}{2} \quad \text{or} \quad y = \frac{-12}{4} = -3.$$

✓ *Check:* $2(\frac{1}{4}) + 5(\frac{1}{2}) - 3 = \frac{1}{2} + \frac{5}{2} - 3 = 0$; $2(9) + 5(-3) - 3 = 18 - 15 - 3 = 0$.

(b) The leading coefficient is already 1:

$$s^2 + s = 3.$$

Half of 1 is $\frac{1}{2}$; add $\frac{1}{4}$:

$$s^2 + s + \frac{1}{4} = 3 + \frac{1}{4} = \frac{13}{4} \Rightarrow \left(s + \frac{1}{2}\right)^2 = \frac{13}{4} \Rightarrow s + \frac{1}{2} = \pm \frac{\sqrt{13}}{2},$$

$$s = \frac{-1 \pm \sqrt{13}}{2} \approx 1.303 \quad \text{or} \quad -2.303.$$

Answer. (a) $y = \frac{1}{2}, -3$ (b) $s = \frac{-1 \pm \sqrt{13}}{2}$

3.5 Linear inequalities

Q.5

Solve the following inequalities. (a) $z + 6 \leq 10 - z$ (b) $3y + 6 \geq 3y - 5$

Solution

(a)

$$z + z \leq 10 - 6 \implies 2z \leq 4 \implies z \leq 2.$$

Dividing by +2, so the sign does not turn. Solution set $(-\infty, 2]$.

(b) Subtract $3y$ from both sides and the variable disappears:

$$3y + 6 \geq 3y - 5 \implies 6 \geq -5.$$

This is a true statement containing no y at all. It is true whatever y is, so *every* real number satisfies the original inequality.

$$\text{Solution set} = \mathbb{R} = (-\infty, \infty).$$

Answer. (a) $z \leq 2$, i.e. $(-\infty, 2]$ (b) all real y ; the inequality reduces to $6 \geq -5$, which is always true.

Where marks are lost

Part (b) is a deliberate trap. “No y left” does not mean “no solution”. If what remains is *true* ($6 \geq -5$) every y works; if what remains is *false* (say $6 \leq -5$) no y works. Read the surviving statement.

3.6 Quadratic inequalities

Q.6

Solve the following second-degree inequalities. (a) $6t^2 + t - 12 > 0$ (b) $2s^2 - 3s - 2 < 0$

Solution

Method for both: find the roots, then decide which side of them the parabola lies on the required side of the axis.

(a) Roots of $6t^2 + t - 12 = 0$: discriminant $= 1 + 288 = 289$, $\sqrt{289} = 17$,

$$t = \frac{-1 \pm 17}{12} \implies t = \frac{16}{12} = \frac{4}{3} \quad \text{or} \quad t = \frac{-18}{12} = -\frac{3}{2}.$$

The coefficient of t^2 is $6 > 0$, so the parabola opens *upwards*: it is below the axis between the roots and above the axis outside them. We want > 0 , i.e. above:

$$t < -\frac{3}{2} \quad \text{or} \quad t > \frac{4}{3}.$$

✓ *Check:* Test $t = 0$ (between the roots): $6(0) + 0 - 12 = -12$, which is not > 0 — correctly excluded.
Test $t = 2$: $24 + 2 - 12 = 14 > 0$ — correctly included.

(b) Roots of $2s^2 - 3s - 2 = 0$: discriminant $= 9 + 16 = 25$,

$$s = \frac{3 \pm 5}{4} \implies s = 2 \quad \text{or} \quad s = -\frac{1}{2}.$$

Coefficient of s^2 is $2 > 0$, so the parabola is below the axis *between* the roots. We want < 0 :

$$-\frac{1}{2} < s < 2.$$

✓ *Check*: Test $s = 0$: $-2 < 0$ — included. Test $s = 3$: $18 - 9 - 2 = 7$, not < 0 — excluded.

Answer. (a) $t < -\frac{3}{2}$ or $t > \frac{4}{3}$ (b) $-\frac{1}{2} < s < 2$

3.7 Absolute value equations and inequalities

Q.7

Solve the following. (a) $|2y + 5| = |y - 4|$ (b) $|y| = |-y + 7|$ (c) $|z^2 - 8| \leq 8$ (d) $|z^2 - 2| \geq 2$

Solution

(a) $|u| = |v|$ splits into two cases, $u = v$ and $u = -v$:

Case 1: $2y + 5 = y - 4 \Rightarrow y = -9$.

Case 2: $2y + 5 = -(y - 4) = -y + 4 \Rightarrow 3y = -1 \Rightarrow y = -\frac{1}{3}$.

✓ *Check*: $y = -9$: $|-18 + 5| = 13$ and $|-9 - 4| = 13$. $y = -\frac{1}{3}$: $|\frac{2}{3} + 5| = \frac{13}{3}$ and $|\frac{1}{3} - 4| = \frac{13}{3}$.

(b) Case 1: $y = -y + 7 \Rightarrow 2y = 7 \Rightarrow y = \frac{7}{2}$.

Case 2: $y = -(-y + 7) = y - 7 \Rightarrow 0 = -7$, a contradiction, so this case yields nothing.

Only $y = \frac{7}{2}$.

(c) $|u| \leq a$ (with $a \geq 0$) is the double inequality $-a \leq u \leq a$:

$$-8 \leq z^2 - 8 \leq 8 \implies 0 \leq z^2 \leq 16.$$

The left half, $z^2 \geq 0$, is automatically true for every real z and imposes nothing. The right half gives

$$z^2 \leq 16 \implies -4 \leq z \leq 4.$$

(d) $|u| \geq a$ splits *outwards*: $u \geq a$ or $u \leq -a$:

$$z^2 - 2 \geq 2 \quad \text{or} \quad z^2 - 2 \leq -2,$$

$$z^2 \geq 4 \quad \text{or} \quad z^2 \leq 0.$$

The first gives $z \leq -2$ or $z \geq 2$. The second, $z^2 \leq 0$, holds only at the single point $z = 0$ — easy to discard by reflex, but it is a genuine part of the solution set.

$$\text{Solution: } z \leq -2, \quad z = 0, \quad \text{or} \quad z \geq 2.$$

✓ *Check*: $z = 0$: $|0 - 2| = 2 \geq 2$. True. $z = 1$: $|1 - 2| = 1 \geq 2$ is false, correctly excluded.

Answer. (a) $y = -9, -\frac{1}{3}$ (b) $y = \frac{7}{2}$ (c) $-4 \leq z \leq 4$ (d) $z \leq -2$ or $z = 0$ or $z \geq 2$

Where marks are lost

$|u| \leq a$ closes *inwards* to a single interval; $|u| \geq a$ opens *outwards* to two. Getting these the wrong way round in (c) and (d) inverts the answer completely.

3.8 Midpoints, distances and a segment length**Q.8, Q.9, Q.10**

Q.8 Find the midpoint of the segment joining (a) (3, 8) and (5, 5); (b) (−2, −4) and (2, 4); (c) (6, 6) and (−3, −3); (d) (0, 9) and (9, 0).

Q.9 Find the distance between (a) (12, 0) and (0, −8); (b) (4, 4) and (−5, −8); (c) (−2, 4) and (1, 0); (d) (−7, −2) and (1, −4).

Q.10 Find the length of the segment joining $C(2, 4)$ and $D(4, 8)$.

Solution

Q.8 — midpoints, $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$:

	Working	Midpoint
(a)	$\left(\frac{3+5}{2}, \frac{8+5}{2} \right)$	(4, 6.5)
(b)	$\left(\frac{-2+2}{2}, \frac{-4+4}{2} \right)$	(0, 0)
(c)	$\left(\frac{6-3}{2}, \frac{6-3}{2} \right)$	(1.5, 1.5)
(d)	$\left(\frac{0+9}{2}, \frac{9+0}{2} \right)$	(4.5, 4.5)

Q.9 — distances, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$:

	Working	Distance
(a)	$\sqrt{(0 - 12)^2 + (-8 - 0)^2} = \sqrt{144 + 64} = \sqrt{208}$	$4\sqrt{13} \approx 14.42$
(b)	$\sqrt{(-5 - 4)^2 + (-8 - 4)^2} = \sqrt{81 + 144} = \sqrt{225}$	15
(c)	$\sqrt{(1 + 2)^2 + (0 - 4)^2} = \sqrt{9 + 16} = \sqrt{25}$	5
(d)	$\sqrt{(1 + 7)^2 + (-4 + 2)^2} = \sqrt{64 + 4} = \sqrt{68}$	$2\sqrt{17} \approx 8.25$

Parts (b) and (c) are the 9–12–15 and 3–4–5 right triangles, which is why they come out whole.

Q.10.

$$|CD| = \sqrt{(4 - 2)^2 + (8 - 4)^2} = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5} \approx 4.47.$$

Answer. Q.8: (4, 6.5), (0, 0), (1.5, 1.5), (4.5, 4.5). Q.9: $4\sqrt{13} \approx 14.42$, 15, 5, $2\sqrt{17} \approx 8.25$.
Q.10: $2\sqrt{5} \approx 4.47$.

Where marks are lost

The distance formula *subtracts* and the midpoint formula *adds*. Mixing them — $(\frac{x_2-x_1}{2}, \dots)$ for a midpoint — is the standard slip, and it silently gives a plausible-looking wrong point.

Linear Equations

Source: *Business Mathematics* 5405/1429, §4.13 SELF ASSESSMENT QUESTIONS, pp. 93–94.

Find the slope, then use one point

$$\begin{array}{ll}
 \text{Slope :} & m = \frac{y_2 - y_1}{x_2 - x_1} \\
 \text{Point-slope :} & y - y_1 = m(x - x_1) \\
 \text{Slope-intercept :} & y = mx + c \\
 \text{Parallel :} & m_1 = m_2 \quad \text{Perpendicular : } m_1 m_2 = -1
 \end{array}$$

Every “find the equation of the line” question in this unit is: get m , then substitute one known point. Nothing else is required.

4.1 Graphing linear equations

Q.1

Graph each of the following linear equations: (a) $9y - 4x = 12$ (b) $y = -x$ (c) $2x + 7y = 14$
 (d) $x = 2y + 3$ (e) $5y - 3x = 13$ (f) $2y = 3x$

Solution

Two points determine a line, and the two cheapest points are the intercepts: set $x = 0$ to get the y -intercept, set $y = 0$ to get the x -intercept.

	Rearranged	x -intercept	y -intercept	Slope m
(a)	$y = \frac{4}{9}x + \frac{4}{3}$	$(-3, 0)$	$(0, \frac{4}{3})$	$\frac{4}{9}$
(b)	$y = -x$	$(0, 0)$	$(0, 0)$	-1
(c)	$y = -\frac{2}{7}x + 2$	$(7, 0)$	$(0, 2)$	$-\frac{2}{7}$
(d)	$y = \frac{1}{2}x - \frac{3}{2}$	$(3, 0)$	$(0, -\frac{3}{2})$	$\frac{1}{2}$
(e)	$y = \frac{3}{5}x + \frac{13}{5}$	$(-\frac{13}{3}, 0)$	$(0, \frac{13}{5})$	$\frac{3}{5}$
(f)	$y = \frac{3}{2}x$	$(0, 0)$	$(0, 0)$	$\frac{3}{2}$

Lines (b) and (f) pass through the origin, so both intercepts coincide there; for those, take a second point such as $x = 1$ — giving $(1, -1)$ and $(1, \frac{3}{2})$ respectively.

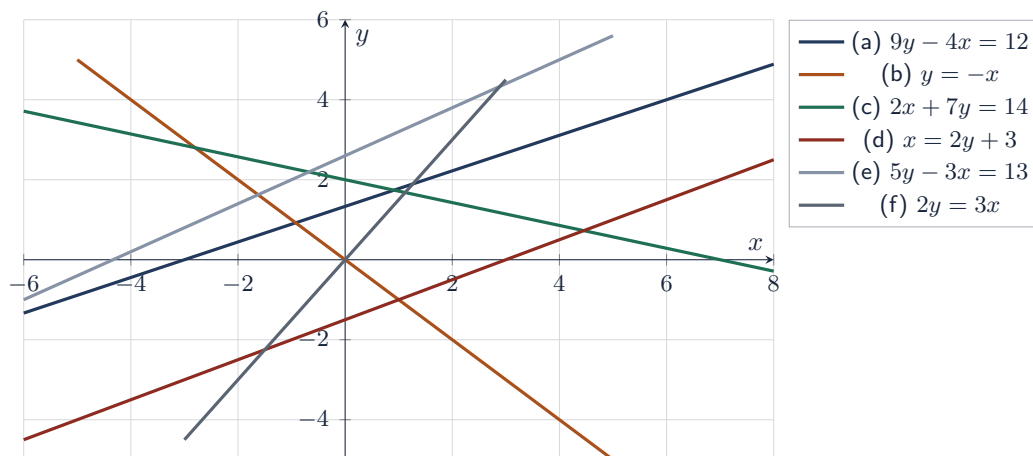


Figure 4.1: The six lines of Q.1. Lines (b) and (f) pass through the origin; the rest are read off their intercepts.

Answer. Each line is plotted from its two intercepts, as tabulated above. Slopes: (a) $\frac{4}{9}$, (b) -1 , (c) $-\frac{2}{7}$, (d) $\frac{1}{2}$, (e) $\frac{3}{5}$, (f) $\frac{3}{2}$.

4.2 Demand, supply and market equilibrium

Q.2

The price and demand for a certain brand of soap are given by $p = 15 - \frac{7}{6}q$, where p is price in dollars and q is demand. (a) Determine the price for a demand of (i) 2, (ii) 5, (iii) 10 units. (b) Determine demand at a price of (i) \$3, (ii) \$10, (iii) \$20. (c) Graph $p = 15 - \frac{7}{6}q$. (d) The price and supply of the soap are given by $p = \frac{2}{3}q$. Determine the supply when the price is (i) \$0, (ii) \$5, (iii) \$30. (e) Graph $p = \frac{2}{3}q$ on the same axes. (f) Determine the equilibrium price and equilibrium quantity.

Solution

(a) **Price from demand.** Substitute q into $p = 15 - \frac{7}{6}q$:

$$q = 2 : p = 15 - \frac{14}{6} = \frac{38}{3} \approx \$12.67; \quad q = 5 : p = 15 - \frac{35}{6} = \frac{55}{6} \approx \$9.17;$$

$$q = 10 : p = 15 - \frac{70}{6} = \frac{10}{3} \approx \$3.33.$$

Price falls as demand rises — the negative slope $-\frac{7}{6}$ is the demand law.

(b) **Demand from price.** Rearrange once and reuse:

$$p = 15 - \frac{7}{6}q \implies \frac{7}{6}q = 15 - p \implies q = \frac{6}{7}(15 - p).$$

$$p = 3 : q = \frac{6}{7}(12) = \frac{72}{7} \approx 10.29; \quad p = 10 : q = \frac{6}{7}(5) = \frac{30}{7} \approx 4.29;$$

$$p = 20 : q = \frac{6}{7}(-5) = -\frac{30}{7} \approx -4.29.$$

A negative quantity is not economically meaningful. It says that at \$20 the price already sits above the demand curve's intercept of \$15, so demand has been choked to zero: the meaningful answer is $q = 0$.

(d) **Supply from price.** $p = \frac{2}{3}q \Rightarrow q = \frac{3}{2}p$:

$$p = 0 : q = 0; \quad p = 5 : q = 7.5; \quad p = 30 : q = 45.$$

Supply rises with price — the positive slope.

(f) **Equilibrium.** The market clears where quantity demanded equals quantity supplied, i.e. where the two lines cross:

$$15 - \frac{7}{6}q = \frac{2}{3}q.$$

Multiply through by 6 to clear fractions:

$$90 - 7q = 4q \Rightarrow 90 = 11q \Rightarrow q^* = \frac{90}{11} \approx 8.18 \text{ units.}$$

$$p^* = \frac{2}{3} \cdot \frac{90}{11} = \frac{60}{11} \approx \$5.45.$$

✓ *Check:* From the demand side: $15 - \frac{7}{6} \cdot \frac{90}{11} = 15 - \frac{105}{11} = \frac{165-105}{11} = \frac{60}{11}$. The two curves agree, so the intersection is right.

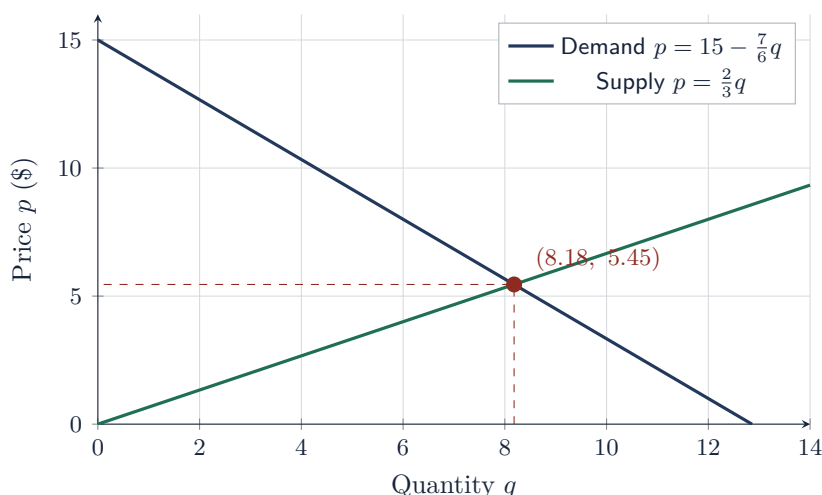


Figure 4.2: Parts (c) and (e) on the same axes. Equilibrium is the intersection: $q^* = 90/11 \approx 8.18$ units at $p^* = 60/11 \approx \$5.45$.

Answer. (a) \$12.67, \$9.17, \$3.33. (b) $72/7 \approx 10.29$, $30/7 \approx 4.29$, and no positive demand at \$20 (the formula returns $-30/7$, so demand is 0). (d) 0, 7.5, 45 units. (f) $q^* = 90/11 \approx 8.18$ units, $p^* = 60/11 \approx \$5.45$.

Where marks are lost

In (b)(iii) a negative quantity is not the answer to write down and move on. State that demand is zero at that price, and say why — the \$20 price is above the demand curve's \$15 vertical intercept.

4.3 Working backwards through a sales tax

Q.3

John paid 7.5% sales tax and \$150 title and license fee when he bought a new car for a total of \$25,868.50. Find the purchase price of the car.

Solution

Let P be the purchase price before tax. The tax is charged on P , and the fixed fee is added afterwards:

$$\underbrace{P}_{\text{price}} + \underbrace{0.075P}_{\text{tax}} + \underbrace{150}_{\text{fee}} = 25,868.50.$$

$$1.075P = 25,868.50 - 150 = 25,718.50 \implies P = \frac{25,718.50}{1.075} = 23,924.19 \text{ (to the nearest cent).}$$

✓ *Check:* $23,924.19 + 0.075(23,924.19) + 150 = 23,924.19 + 1,794.31 + 150 = 25,868.50$.

Answer. The purchase price of the car was approximately **\$23,924.19**.

Where marks are lost

Do not subtract 7.5% of the *total*. $25,718.50 \times 0.925 = 23,789.61$ is wrong, because the tax was 7.5% of the smaller pre-tax price, not of the larger total. Always divide by $1 + r$; never multiply by $1 - r$.

4.4 Slopes through pairs of points

Q.4

Find the slope of the straight line passing through the indicated points. (i) $(4, 1), (7, 5)$ (ii) $(5, -3), (6, -4)$ (iii) $(1, 0), (0, 5)$ (iv) $(2, -4), (3, -4)$

Solution

$m = \frac{y_2 - y_1}{x_2 - x_1}$ throughout.

	Working	m	Reading
(i)	$\frac{5-1}{7-4} = \frac{4}{3}$	$\frac{4}{3}$	rises left to right
(ii)	$\frac{-4-(-3)}{6-5} = \frac{-1}{1}$	-1	falls at 45°
(iii)	$\frac{5-0}{0-1} = \frac{5}{-1}$	-5	steeply falling
(iv)	$\frac{-4-(-4)}{3-2} = \frac{0}{1}$	0	horizontal line $y = -4$

Answer. (i) $\frac{4}{3}$ (ii) -1 (iii) -5 (iv) 0

Where marks are lost

Keep the two points in the same order in numerator and denominator. Computing $\frac{y_2 - y_1}{x_1 - x_2}$ flips the sign of the slope, which then propagates into every part of a later question.

4.5 Equations of lines from stated properties**Q.5**

Find the equation of the line with the indicated properties and sketch it. **(a)** passing through $(-1, 3)$ and parallel to $y = 4x - 5$; **(b)** slope 0 and y -intercept $-\frac{1}{2}$; **(c)** passing through $(3, -1)$ and $(-2, -9)$; **(d)** passing through the origin with slope -5 ; **(e)** passing through $(-5, 4)$ and perpendicular to $2y = x + 1$; **(f)** perpendicular to $y = 3x - 5$ and passing through $(3, 4)$; **(g)** passing through $(2, -8)$ and parallel to $x = -4$.

Solution

(a) Parallel lines share a slope, so $m = 4$. Point-slope through $(-1, 3)$:

$$y - 3 = 4(x + 1) \implies y = 4x + 7.$$

(b) Slope 0 means a horizontal line; its equation is just its y -intercept:

$$y = -\frac{1}{2}.$$

(c) First the slope:

$$m = \frac{-9 - (-1)}{-2 - 3} = \frac{-8}{-5} = \frac{8}{5}.$$

Then through $(3, -1)$:

$$y + 1 = \frac{8}{5}(x - 3) \implies 5y + 5 = 8x - 24 \implies 5y = 8x - 29.$$

✓ *Check:* At $x = -2$: $5y = -16 - 29 = -45$, so $y = -9$. The second point is on it.

(d) Through the origin, so $c = 0$:

$$y = -5x.$$

(e) Write $2y = x + 1$ as $y = \frac{1}{2}x + \frac{1}{2}$, so its slope is $\frac{1}{2}$. The perpendicular slope is the negative reciprocal:

$$m = -\frac{1}{1/2} = -2.$$

Through $(-5, 4)$:

$$y - 4 = -2(x + 5) \implies y = -2x - 6.$$

(f) $y = 3x - 5$ has $m = 3$, so the perpendicular slope is $-\frac{1}{3}$. Through $(3, 4)$:

$$y - 4 = -\frac{1}{3}(x - 3) \implies 3y - 12 = -x + 3 \implies x + 3y = 15.$$

(g) $x = -4$ is a *vertical* line, which has no slope at all. A line parallel to it is also vertical, and passes through $x = 2$:

$$x = 2.$$

Equation	Form
(a) $y = 4x + 7$	slope 4, y -intercept 7
(b) $y = -\frac{1}{2}$	horizontal
(c) $5y = 8x - 29$	slope $\frac{8}{5}$
(d) $y = -5x$	through the origin
(e) $y = -2x - 6$	slope -2
(f) $x + 3y = 15$	slope $-\frac{1}{3}$
(g) $x = 2$	vertical, undefined slope

Answer. (a) $y = 4x + 7$ (b) $y = -\frac{1}{2}$ (c) $5y = 8x - 29$ (d) $y = -5x$ (e) $y = -2x - 6$
 (f) $x + 3y = 15$ (g) $x = 2$

Where marks are lost

Part (g) catches most candidates. The rule $m_1 m_2 = -1$ does not apply to vertical lines, because a vertical line has no slope. A line parallel to $x = -4$ is $x = \text{constant}$, and the constant comes from the x -coordinate of the given point, not the y .

4.6 A linear cost function

Q.6

A small company produces chairs. The weekly fixed cost is \$1,100 and the variable cost is \$40 per chair. Find the total weekly cost of manufacturing x chairs. How many chairs can be manufactured for a total weekly cost of \$4,500?

Solution

A linear cost function is

$$C(x) = \underbrace{\text{variable cost per unit} \times x}_{40} + \underbrace{\text{fixed cost}}_{1100},$$

so

$$C(x) = 40x + 1100.$$

Note the structure: the fixed cost is the y -intercept (the cost of producing nothing), the variable cost is the slope.

Setting $C(x) = 4500$:

$$40x + 1100 = 4500 \implies 40x = 3400 \implies x = 85.$$

✓ *Check:* $40(85) + 1100 = 3400 + 1100 = 4500$.

Answer. $C(x) = 40x + 1100$; a budget of \$4,500 buys **85 chairs** per week.

4.7 Linear interpolation of sales

Q.7

The sales of a company were \$33,000 in its third year of operation and \$95,000 in its sixth year. If y denotes sales in year x , what were the sales in the fifth year?

Solution

“ y denotes sales in year x ” with two data points means we are asked to assume sales grow *linearly*. The two points are $(3, 33000)$ and $(6, 95000)$.

Slope — the annual increase in sales:

$$m = \frac{95000 - 33000}{6 - 3} = \frac{62000}{3} \approx 20,666.67 \text{ dollars per year.}$$

Point-slope through $(3, 33000)$:

$$y - 33000 = \frac{62000}{3}(x - 3).$$

At $x = 5$:

$$y = 33000 + \frac{62000}{3}(2) = 33000 + \frac{124000}{3} = 33000 + 41,333.33 = 74,333.33.$$

✓ *Check:* At $x = 6$ the formula gives $33000 + \frac{62000}{3}(3) = 33000 + 62000 = 95,000$, matching the stated sixth-year figure.

Answer. Sales in the fifth year were approximately **\$74,333.33**, on the linear model $y = 33000 + \frac{62000}{3}(x - 3)$.

Where marks are lost

The slope is the change *per year*, so the denominator is $6 - 3 = 3$, not 6 and not 2. Dividing 62,000 by the wrong span is the only real difficulty in this question.

Matrices

Source: *Business Mathematics* 5405/1429, §5.8 SELF ASSESSMENT QUESTIONS, pp. 107–108.

Conformability, and the one rule that is not like arithmetic

$A_{m \times n} B_{n \times p}$ is defined only when the *inner* dimensions agree, and the product is $m \times p$. Entry (i, j) of AB is row i of A dotted with column j of B .

Addition and scalar multiplication behave like arithmetic. Multiplication does not: in general $AB \neq BA$. The transpose rules follow from this:

$$(A + B)^t = A^t + B^t, \quad (A^t)^t = A, \quad \boxed{(AB)^t = B^t A^t}$$

— note the reversal in the last one.

5.1 Transposes

Q.1

Find the transpose of each of the following matrices:

$$(i) A = \begin{pmatrix} -1 & 0 & 0 \\ 6 & 9 & -2 \\ -4 & 5 & 1 \end{pmatrix} \quad (ii) B = \begin{pmatrix} 6 & -6 \\ 4 & 9 \\ -2 & 0 \end{pmatrix} \quad (iii) C = \begin{pmatrix} 0 & 0 & 2 \end{pmatrix}$$

Solution

The transpose turns rows into columns: $(A^t)_{ij} = A_{ji}$. A matrix of order $m \times n$ transposes to one of order $n \times m$.

$$(i) A^t = \begin{pmatrix} -1 & 6 & -4 \\ 0 & 9 & 5 \\ 0 & -2 & 1 \end{pmatrix} \quad (ii) B^t = \begin{pmatrix} 6 & 4 & -2 \\ -6 & 9 & 0 \end{pmatrix} \quad (iii) C^t = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

Orders: A is 3×3 so A^t is 3×3 ; B is 3×2 so B^t is 2×3 ; C is a 1×3 row vector, so C^t is a 3×1 column vector.

Answer. As displayed above. Note that transposing reverses the order: $3 \times 2 \rightarrow 2 \times 3$.

5.2 Verifying the double transpose

Q.2

If $A = \begin{pmatrix} -8 & 0 & 7 \\ 5 & 2 & -1 \\ -4 & 0 & 3 \end{pmatrix}$, verify that $(A^t)^t = A$.

Solution

Transpose once — rows of A become columns:

$$A^t = \begin{pmatrix} -8 & 5 & -4 \\ 0 & 2 & 0 \\ 7 & -1 & 3 \end{pmatrix}.$$

Transpose again — rows of A^t become columns:

$$(A^t)^t = \begin{pmatrix} -8 & 0 & 7 \\ 5 & 2 & -1 \\ -4 & 0 & 3 \end{pmatrix} = A. \quad \checkmark$$

Every entry returns to its starting position, because $((A^t)^t)_{ij} = (A^t)_{ji} = A_{ij}$.

Answer. $(A^t)^t = A$, verified entry by entry.

5.3 Matrix algebra with two 3×3 matrices

Q.3

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & 3 \\ 3 & 4 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & -1 & 1 \\ 1 & 4 & 3 \\ 3 & -3 & -2 \end{pmatrix}$. Find (i) $4A - 3B$, (ii) $A + 2(B - A)$, (iii) $2A^t + 3B^t$, (iv) AB ; and verify that $(A + B)^t = A^t + B^t$ and $(AB)^t = B^t A^t$.

Solution

(i) $4A - 3B$. Scalar multiplication and subtraction are entrywise:

$$4A = \begin{pmatrix} 4 & 8 & 12 \\ -4 & -8 & 12 \\ 12 & 16 & 20 \end{pmatrix}, \quad 3B = \begin{pmatrix} 12 & -3 & 3 \\ 3 & 12 & 9 \\ 9 & -9 & -6 \end{pmatrix},$$

$$4A - 3B = \begin{pmatrix} 4 - 12 & 8 + 3 & 12 - 3 \\ -4 - 3 & -8 - 12 & 12 - 9 \\ 12 - 9 & 16 + 9 & 20 + 6 \end{pmatrix} = \begin{pmatrix} -8 & 11 & 9 \\ -7 & -20 & 3 \\ 3 & 25 & 26 \end{pmatrix}.$$

(ii) $A + 2(B - A)$. Simplify algebraically before touching numbers:

$$A + 2(B - A) = A + 2B - 2A = 2B - A.$$

$$2B = \begin{pmatrix} 8 & -2 & 2 \\ 2 & 8 & 6 \\ 6 & -6 & -4 \end{pmatrix}, \quad 2B - A = \begin{pmatrix} 8-1 & -2-2 & 2-3 \\ 2+1 & 8+2 & 6-3 \\ 6-3 & -6-4 & -4-5 \end{pmatrix} = \begin{pmatrix} 7 & -4 & -1 \\ 3 & 10 & 3 \\ 3 & -10 & -9 \end{pmatrix}.$$

(iii) $2A^t + 3B^t$.

$$A^t = \begin{pmatrix} 1 & -1 & 3 \\ 2 & -2 & 4 \\ 3 & 3 & 5 \end{pmatrix}, \quad B^t = \begin{pmatrix} 4 & 1 & 3 \\ -1 & 4 & -3 \\ 1 & 3 & -2 \end{pmatrix},$$

$$2A^t + 3B^t = \begin{pmatrix} 2 & -2 & 6 \\ 4 & -4 & 8 \\ 6 & 6 & 10 \end{pmatrix} + \begin{pmatrix} 12 & 3 & 9 \\ -3 & 12 & -9 \\ 3 & 9 & -6 \end{pmatrix} = \begin{pmatrix} 14 & 1 & 15 \\ 1 & 8 & -1 \\ 9 & 15 & 4 \end{pmatrix}.$$

(iv) AB . Row of A against column of B . Taking the first row (1, 2, 3) against each column of B :

$$\begin{aligned} (1)(4) + (2)(1) + (3)(3) &= 4 + 2 + 9 = 15, \\ (1)(-1) + (2)(4) + (3)(-3) &= -1 + 8 - 9 = -2, \\ (1)(1) + (2)(3) + (3)(-2) &= 1 + 6 - 6 = 1. \end{aligned}$$

Repeating for rows 2 and 3:

$$AB = \begin{pmatrix} 15 & -2 & 1 \\ 3 & -16 & -13 \\ 31 & -2 & 5 \end{pmatrix}.$$

Verification 1: $(A + B)^t = A^t + B^t$.

$$A + B = \begin{pmatrix} 5 & 1 & 4 \\ 0 & 2 & 6 \\ 6 & 1 & 3 \end{pmatrix} \implies (A + B)^t = \begin{pmatrix} 5 & 0 & 6 \\ 1 & 2 & 1 \\ 4 & 6 & 3 \end{pmatrix},$$

$$A^t + B^t = \begin{pmatrix} 1 & -1 & 3 \\ 2 & -2 & 4 \\ 3 & 3 & 5 \end{pmatrix} + \begin{pmatrix} 4 & 1 & 3 \\ -1 & 4 & -3 \\ 1 & 3 & -2 \end{pmatrix} = \begin{pmatrix} 5 & 0 & 6 \\ 1 & 2 & 1 \\ 4 & 6 & 3 \end{pmatrix}. \quad \checkmark$$

Verification 2: $(AB)^t = B^t A^t$.

$$(AB)^t = \begin{pmatrix} 15 & 3 & 31 \\ -2 & -16 & -2 \\ 1 & -13 & 5 \end{pmatrix}.$$

Computing $B^t A^t$ — first row of B^t is (4, 1, 3), first column of A^t is (1, 2, 3)^t, giving $4 + 2 + 9 = 15$, and so on:

$$B^t A^t = \begin{pmatrix} 15 & 3 & 31 \\ -2 & -16 & -2 \\ 1 & -13 & 5 \end{pmatrix}. \quad \checkmark$$

The order genuinely matters: $A^t B^t$ is a different matrix entirely.

Answer. (i) $\begin{pmatrix} -8 & 11 & 9 \\ -7 & -20 & 3 \\ 3 & 25 & 26 \end{pmatrix}$ (ii) $\begin{pmatrix} 7 & -4 & -1 \\ 3 & 10 & 3 \\ 3 & -10 & -9 \end{pmatrix}$ (iii) $\begin{pmatrix} 14 & 1 & 15 \\ 1 & 8 & -1 \\ 9 & 15 & 4 \end{pmatrix}$ (iv) $AB = \begin{pmatrix} 15 & -2 & 1 \\ 3 & -16 & -13 \\ 31 & -2 & 5 \end{pmatrix}$; both transpose identities verified.

Where marks are lost

$(AB)^t = B^t A^t$, **not** $A^t B^t$. The reversal is not a convention — it is forced by conformability, and writing it the wrong way round gives a matrix that is usually not even the right shape.

5.4 A commuting-matrices proof

Q.4

Prove that if $AB = BA$, then $(A + B)(A - B) = A^2 - B^2$.

Solution

Expand the left side using the distributive law, which *does* hold for matrices. What does not hold in general is commutativity, so keep every product in the order it arises:

$$(A + B)(A - B) = A(A - B) + B(A - B) = A^2 - AB + BA - B^2.$$

So far this is true for any A and B . Now apply the hypothesis $AB = BA$, which lets the middle two terms cancel:

$$-AB + BA = -AB + AB = 0.$$

Therefore

$$(A + B)(A - B) = A^2 - B^2. \quad \blacksquare$$

The converse is worth noting. Running the argument backwards, if $(A+B)(A-B) = A^2 - B^2$ then $BA - AB = 0$, so A and B must commute. The familiar difference-of-squares identity holds for matrices *exactly* when $AB = BA$ — which is why the hypothesis is stated.

Answer. Expanding gives $A^2 - AB + BA - B^2$; the hypothesis $AB = BA$ makes $-AB + BA = 0$, leaving $A^2 - B^2$.

5.5 Valuing an inventory by matrix multiplication

Q.5

A pet store has 6 kittens, 8 parrots and 12 puppies. If each kitten costs \$28, each parrot \$30 and each puppy \$40, use matrix multiplication to find the total value of the store's inventory.

Solution

Write the quantities as a 1×3 row and the prices as a 3×1 column. The inner dimensions agree ($3 = 3$), so the product is 1×1 — a single number, which is exactly what a total value should be.

$$Q = (6 \quad 8 \quad 12), \quad P = \begin{pmatrix} 28 \\ 30 \\ 40 \end{pmatrix},$$

$$QP = (6 \quad 8 \quad 12) \begin{pmatrix} 28 \\ 30 \\ 40 \end{pmatrix} = (6)(28) + (8)(30) + (12)(40)$$

$$= 168 + 240 + 480 = (888).$$

Answer. The inventory is worth **\$888**.

Where marks are lost

Order the factors so the inner dimensions match. PQ with P as 3×1 and Q as 1×3 is defined, but produces a 3×3 matrix of every price-times-quantity cross-product — not a total.

5.6 Match-day revenue

Q.6

Attendance for the first three football games of the season is given below. Adult tickets sold for \$3.00 and student tickets for \$1.50. Use matrix multiplication to find the revenue for each game.

	Adults	Students
Game 1	340	180
Game 2	250	195
Game 3	300	220

Solution

Attendance is a 3×2 matrix (one row per game), prices a 2×1 column. Inner dimensions agree, so the product is 3×1 — one revenue figure per game.

$$A = \begin{pmatrix} 340 & 180 \\ 250 & 195 \\ 300 & 220 \end{pmatrix}, \quad P = \begin{pmatrix} 3.00 \\ 1.50 \end{pmatrix},$$

$$AP = \begin{pmatrix} 340(3.00) + 180(1.50) \\ 250(3.00) + 195(1.50) \\ 300(3.00) + 220(1.50) \end{pmatrix} = \begin{pmatrix} 1020 + 270 \\ 750 + 292.50 \\ 900 + 330 \end{pmatrix} = \begin{pmatrix} 1290.00 \\ 1042.50 \\ 1230.00 \end{pmatrix}.$$

	Adults @ \$3	Students @ \$1.50	Revenue
Game 1	\$1,020	\$270.00	\$1,290.00
Game 2	\$750	\$292.50	\$1,042.50
Game 3	\$900	\$330.00	\$1,230.00
Season			\$3,562.50

Answer. Game 1: \$1,290.00; Game 2: \$1,042.50; Game 3: \$1,230.00. Total for the three games: \$3,562.50.

5.7 A scalar for a price rise

Q.7

Let $A = \begin{pmatrix} a_1 & a_2 & a_3 \end{pmatrix}$ represent the prices of products X, Y, Z . If prices are to be increased by 15%, by what scalar can A be multiplied to obtain the new prices?

Solution

A 15% increase means each new price is the old price *plus* 15% of it:

$$a_i^{\text{new}} = a_i + 0.15a_i = 1.15a_i.$$

Because the same factor applies to every entry, this is exactly scalar multiplication:

$$A^{\text{new}} = 1.15A = \begin{pmatrix} 1.15a_1 & 1.15a_2 & 1.15a_3 \end{pmatrix}.$$

Answer. Multiply by the scalar **1.15** (equivalently $\frac{23}{20}$).

Where marks are lost

The scalar is 1.15, not 0.15. Multiplying by 0.15 gives the size of the *increase* alone and discards the original price.

5.8 Burger Barn: three branches, three products

Q.8

Burger Barn's three locations sell fries, burgers and soft drinks. Branch I sells 500 orders of fries, 150 burgers and 300 soft drinks daily. Branches II and III sell 900 and 700 orders of fries respectively per day. Branch II sells 440 burgers and 830 soft drinks, while Branch III sells 580 burgers and 1200 soft drinks each day. **(a)** Write a 3×3 matrix representing daily sales for all locations. **(b)** If fries cost \$0.92 per order, burgers \$1.54 each and soft drinks \$0.58 each, write a 1×3 matrix representing the prices. **(c)** What is the total daily income from all the locations?

Solution

(a) The sales matrix. Rows are branches, columns are products:

$$S = \begin{matrix} & \begin{matrix} \text{fries} & \text{burgers} & \text{drinks} \end{matrix} \\ \begin{matrix} \text{I} \\ \text{II} \\ \text{III} \end{matrix} & \begin{pmatrix} 500 & 150 & 300 \\ 900 & 440 & 830 \\ 700 & 580 & 1200 \end{pmatrix} \end{matrix}$$

(b) The price matrix. As given, a 1×3 row:

$$P = (0.92 \quad 1.54 \quad 0.58).$$

(c) Total daily income. S is 3×3 and P is 1×3 , so SP is *not* conformable. Transpose the prices into a column first, and the inner dimensions agree:

$$SP^t = \begin{pmatrix} 500 & 150 & 300 \\ 900 & 440 & 830 \\ 700 & 580 & 1200 \end{pmatrix} \begin{pmatrix} 0.92 \\ 1.54 \\ 0.58 \end{pmatrix}.$$

Branch by branch:

$$\text{I : } 500(0.92) + 150(1.54) + 300(0.58) = 460 + 231 + 174 = 865.00,$$

$$\text{II : } 900(0.92) + 440(1.54) + 830(0.58) = 828 + 677.60 + 481.40 = 1987.00,$$

$$\text{III : } 700(0.92) + 580(1.54) + 1200(0.58) = 644 + 893.20 + 696 = 2233.20.$$

$$SP^t = \begin{pmatrix} 865.00 \\ 1987.00 \\ 2233.20 \end{pmatrix}.$$

Total across all branches:

$$865.00 + 1987.00 + 2233.20 = 5085.20.$$

Branch	Fries	Burgers	Drinks	Daily income
I	500	150	300	\$865.00
II	900	440	830	\$1,987.00
III	700	580	1200	\$2,233.20
Total				\$5,085.20

Answer. (a) $S = \begin{pmatrix} 500 & 150 & 300 \\ 900 & 440 & 830 \\ 700 & 580 & 1200 \end{pmatrix}$ (b) $P = \begin{pmatrix} 0.92 & 1.54 & 0.58 \end{pmatrix}$ (c) Branch incomes \$865.00, \$1,987.00 and \$2,233.20, giving a total daily income of **\$5,085.20**.

Where marks are lost

The question deliberately asks for P as a *row*, then asks for a total that requires it as a *column*. Say so in your working: “ SP is not conformable, so compute SP^t .” Examiners award that observation.

Determinants and Inverses

Source: *Business Mathematics* 5405/1429, §6.8 SELF ASSESSMENT QUESTIONS, pp. 131–132.

Cofactors, inverse, Cramer

For a 3×3 matrix, the cofactor of entry a_{ij} is $C_{ij} = (-1)^{i+j} M_{ij}$, where M_{ij} is the determinant left after deleting row i and column j . The sign pattern is

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}.$$

Then

$$\det A = \sum_j a_{ij} C_{ij} \text{ (any row)}, \quad A^{-1} = \frac{1}{\det A} \text{adj } A, \quad \text{adj } A = [C_{ij}]^t.$$

Cramer's rule: for $Ax = b$, $x_k = \frac{D_k}{D}$, where $D = \det A$ and D_k replaces column k of A by b . It requires $D \neq 0$.

Four shortcuts that turn a long determinant into one line

- Expand along the row or column with the *most zeros*.
- A common factor in a whole row or column comes outside the determinant.
- If two rows (or columns) are *identical* or *proportional*, $\det = 0$.
- Adding a multiple of one row to another leaves the determinant unchanged — use it to manufacture zeros.

6.1 Evaluating determinants

Q.1

Evaluate the following determinants, using properties of determinants where possible.

$$\begin{array}{lll} \text{(a)} \begin{vmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{vmatrix} & \text{(b)} \begin{vmatrix} 2a & a & a \\ b & 2b & b \\ c & c & 2c \end{vmatrix} & \text{(c)} \begin{vmatrix} 1 & 2 & 3x \\ 2 & 3 & 6x \\ 3 & 5 & 9x \end{vmatrix} \end{array}$$

$$\begin{array}{lll} \text{(d)} \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} & \text{(e)} \begin{vmatrix} 2 & 3 & -1 & 1 \\ 3 & 0 & 2 & 5 \\ 0 & 8 & 6 & 3 \\ 7 & -4 & 5 & 0 \end{vmatrix} & \text{(f)} \begin{vmatrix} 1 & 0 & -2 & 3 \\ 0 & 0 & 0 & -5 \\ 4 & 2 & 1 & 4 \\ 1 & 5 & -4 & 3 \end{vmatrix} \end{array}$$

Solution

(a) Expand along the second row, which contains a zero:

$$\begin{vmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{vmatrix} = 2 \begin{vmatrix} 1 & 0 \\ -3 & 5 \end{vmatrix} - 3 \begin{vmatrix} 1 & 0 \\ 2 & 5 \end{vmatrix} + (-1) \begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix} \\ = 2(5 - 0) - 3(5 - 0) - 1(-3 - 2) \\ = 10 - 15 + 5 = 0.$$

A zero determinant means the rows are linearly dependent — indeed $R_3 = 3R_2 - R_1$.

(b) Factor a from row 1, b from row 2, c from row 3:

$$\begin{vmatrix} 2a & a & a \\ b & 2b & b \\ c & c & 2c \end{vmatrix} = abc \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix}.$$

$$\begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} = 2(4 - 1) - 1(2 - 1) + 1(1 - 2) = 6 - 1 - 1 = 4.$$

Value = $4abc$.

(c) The third column is $3x$ times $(1, 2, 3)^t$. Factor $3x$ out:

$$\begin{vmatrix} 1 & 2 & 3x \\ 2 & 3 & 6x \\ 3 & 5 & 9x \end{vmatrix} = 3x \begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 3 & 5 & 3 \end{vmatrix}.$$

Columns 1 and 3 are now identical, so the determinant is 0:

$$\text{Value} = 3x \cdot 0 = 0.$$

(d) Each row sums to zero: $(a - b) + (b - c) + (c - a) = 0$, and likewise for rows 2 and 3. So adding columns 2 and 3 to column 1 — an operation that does not change the determinant — makes column 1 entirely zero:

$$C_1 \rightarrow C_1 + C_2 + C_3 : \begin{vmatrix} 0 & b - c & c - a \\ 0 & c - a & a - b \\ 0 & a - b & b - c \end{vmatrix} = 0.$$

Value = 0.

(e) A 4×4 . Expand along row 3, $(0, 8, 6, 3)$, whose leading zero kills one term. With signs $(-1)^{3+j}$ giving $-, +, -, +$ across the row, so that positions 2 and 4 carry a minus:

$$\det = -8M_{32} + 6M_{33} - 3M_{34},$$

$$M_{32} = \begin{vmatrix} 2 & -1 & 1 \\ 3 & 2 & 5 \\ 7 & 5 & 0 \end{vmatrix} = 2(0 - 25) + 1(0 - 35) + 1(15 - 14) = -50 - 35 + 1 = -84,$$

$$M_{33} = \begin{vmatrix} 2 & 3 & 1 \\ 3 & 0 & 5 \\ 7 & -4 & 0 \end{vmatrix} = 2(0 + 20) - 3(0 - 35) + 1(-12 - 0) = 40 + 105 - 12 = 133,$$

$$M_{34} = \begin{vmatrix} 2 & 3 & -1 \\ 3 & 0 & 2 \\ 7 & -4 & 5 \end{vmatrix} = 2(0 + 8) - 3(15 - 14) - 1(-12 - 0) = 16 - 3 + 12 = 25.$$

$$\det = -8(-84) + 6(133) - 3(25) = 672 + 798 - 75 = 1395.$$

✓ *Check:* Expanding instead along column 1 gives $2(296) + 3(-31) + 7(128) = 592 - 93 + 896 = 1395$. Two independent expansions agree.

(f) Row 2 is $(0, 0, 0, -5)$ — three zeros, so expand there and only one term survives. The sign at position $(2, 4)$ is $(-1)^{2+4} = +$:

$$\det = (-5) M_{24}, \quad M_{24} = \begin{vmatrix} 1 & 0 & -2 \\ 4 & 2 & 1 \\ 1 & 5 & -4 \end{vmatrix}.$$

$$M_{24} = 1(2(-4) - 1(5)) - 0 + (-2)(4(5) - 2(1)) = 1(-13) - 2(18) = -13 - 36 = -49.$$

$$\det = (-5)(-49) = 245.$$

Answer. (a) 0 (b) $4abc$ (c) 0 (d) 0 (e) 1395 (f) 245

Where marks are lost

Parts (a), (c) and (d) are all zero, and each for a different structural reason — dependent rows, a repeated column after factoring, rows summing to zero. Recognising the structure takes one line; brute-force expansion takes ten and usually goes wrong. Look for the pattern before you expand.

6.2 Inverses, and whether they distribute over a product

Q.2

Let $A = \begin{pmatrix} 1 & 3 & -1 \\ 5 & 2 & 3 \\ -2 & 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & -2 & 0 \\ 3 & 5 & 1 \\ -2 & -3 & 4 \end{pmatrix}$. (a) Find A^{-1} , B^{-1} , $A^{-1}B^{-1}$, $B^{-1}A^{-1}$ and

$(AB)^{-1}$. (b) Is it true that $(AB)^{-1} = A^{-1}B^{-1}$? (c) Is it true that $(AB)^{-1} = B^{-1}A^{-1}$? (d) Does $A + B$ have an inverse? If so, what is it? (e) Does $A - B$ have an inverse? If so, what is it?

Solution

(a) A^{-1} . First the determinant, expanding along row 1:

$$\det A = 1(0 - 3) - 3(0 + 6) + (-1)(5 + 4) = -3 - 18 - 9 = -30 \neq 0.$$

Cofactors of A , with the $+, -, +$ sign pattern applied:

$$\begin{aligned}C_{11} &= -3, & C_{12} &= -6, & C_{13} &= 9, \\C_{21} &= -1, & C_{22} &= -2, & C_{23} &= -7, \\C_{31} &= 11, & C_{32} &= -8, & C_{33} &= -13.\end{aligned}$$

The adjoint is the *transpose* of the cofactor matrix:

$$\text{adj } A = \begin{pmatrix} -3 & -1 & 11 \\ -6 & -2 & -8 \\ 9 & -7 & -13 \end{pmatrix}, \quad A^{-1} = \frac{1}{-30} \begin{pmatrix} -3 & -1 & 11 \\ -6 & -2 & -8 \\ 9 & -7 & -13 \end{pmatrix} = \frac{1}{30} \begin{pmatrix} 3 & 1 & -11 \\ 6 & 2 & 8 \\ -9 & 7 & 13 \end{pmatrix}.$$

✓ *Check:* Row 1 of A times column 1 of A^{-1} : $\frac{1}{30}(1(3) + 3(6) + (-1)(-9)) = \frac{30}{30} = 1$. Row 1 times column 3: $\frac{1}{30}(-11 + 24 - 13) = 0$. Correct.

B^{-1} .

$$\det B = -1(20 + 3) + 2(12 + 2) + 0 = -23 + 28 = 5 \neq 0.$$

$$\text{adj } B = \begin{pmatrix} 23 & 8 & -2 \\ -14 & -4 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad B^{-1} = \frac{1}{5} \begin{pmatrix} 23 & 8 & -2 \\ -14 & -4 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

✓ *Check:* Row 1 of B times column 1 of B^{-1} : $\frac{1}{5}(-1(23) + (-2)(-14) + 0(1)) = \frac{5}{5} = 1$.

AB and its inverse.

$$AB = \begin{pmatrix} 10 & 16 & -1 \\ -5 & -9 & 14 \\ 5 & 9 & 1 \end{pmatrix}, \quad \det(AB) = \det A \cdot \det B = (-30)(5) = -150.$$

$$B^{-1}A^{-1} = \frac{1}{5} \begin{pmatrix} 23 & 8 & -2 \\ -14 & -4 & 1 \\ 1 & 1 & 1 \end{pmatrix} \cdot \frac{1}{-30} \begin{pmatrix} -3 & -1 & 11 \\ -6 & -2 & -8 \\ 9 & -7 & -13 \end{pmatrix} = \frac{1}{-150} \begin{pmatrix} -135 & -25 & 215 \\ 75 & 15 & -135 \\ 0 & -10 & -10 \end{pmatrix},$$

$$B^{-1}A^{-1} = \begin{pmatrix} \frac{9}{10} & \frac{1}{6} & -\frac{43}{30} \\ -\frac{1}{2} & -\frac{1}{10} & \frac{9}{10} \\ 0 & \frac{1}{15} & \frac{1}{15} \end{pmatrix} = (AB)^{-1}.$$

✓ *Check:* Row 1 of AB times column 1 of this matrix: $10(0.9) + 16(-0.5) + (-1)(0) = 9 - 8 = 1$. Row 1 times column 2: $\frac{10}{6} - \frac{16}{10} - \frac{1}{15} = 0$. It is genuinely the inverse of AB .

By contrast, the *other* order gives

$$A^{-1}B^{-1} = \frac{1}{-150} \begin{pmatrix} -44 & \dots & \dots \\ \dots & \dots & \dots \\ \dots & \dots & \dots \end{pmatrix} \quad \text{with } (1,1) \text{ entry } \frac{-44}{-150} = \frac{22}{75} \approx 0.293,$$

which is not the 0.9 required in position $(1,1)$ of $(AB)^{-1}$.

(b) No. $A^{-1}B^{-1} \neq (AB)^{-1}$, as the entry above shows. The two agree only when A and B commute.

(c) **Yes.** $(AB)^{-1} = B^{-1}A^{-1}$ always, whenever both inverses exist. The one-line proof:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I.$$

The order reverses for the same reason it does for transposes.

(d) **Does $A + B$ have an inverse?**

$$A+B = \begin{pmatrix} 0 & 1 & -1 \\ 8 & 7 & 4 \\ -4 & -2 & 4 \end{pmatrix}, \quad \det(A+B) = 0 - 1(32+16) - 1(-16+28) = -48 - 12 = -60 \neq 0,$$

$$\text{so yes. With } \text{adj}(A+B) = \begin{pmatrix} 36 & -2 & 11 \\ -48 & -4 & -8 \\ 12 & -4 & -8 \end{pmatrix},$$

$$(A+B)^{-1} = \frac{1}{-60} \begin{pmatrix} 36 & -2 & 11 \\ -48 & -4 & -8 \\ 12 & -4 & -8 \end{pmatrix} = \begin{pmatrix} -\frac{3}{5} & \frac{1}{30} & -\frac{11}{60} \\ \frac{4}{5} & \frac{1}{15} & \frac{2}{15} \\ -\frac{1}{5} & \frac{1}{15} & \frac{2}{15} \end{pmatrix}.$$

✓ *Check:* Row 2 of $A+B$ times column 2: $\frac{8}{30} + \frac{7}{15} + \frac{4}{15} = \frac{4+7+4}{15} = 1$.

(e) **Does $A - B$ have an inverse?**

$$A-B = \begin{pmatrix} 2 & 5 & -1 \\ 2 & -3 & 2 \\ 0 & 4 & -4 \end{pmatrix}, \quad \det(A-B) = 2(12-8) - 5(-8-0) - 1(8-0) = 8 + 40 - 8 = 40 \neq 0,$$

$$\text{so yes. With } \text{adj}(A-B) = \begin{pmatrix} 4 & 16 & 7 \\ 8 & -8 & -6 \\ 8 & -8 & -16 \end{pmatrix},$$

$$(A-B)^{-1} = \frac{1}{40} \begin{pmatrix} 4 & 16 & 7 \\ 8 & -8 & -6 \\ 8 & -8 & -16 \end{pmatrix} = \begin{pmatrix} \frac{1}{10} & \frac{2}{5} & \frac{7}{40} \\ \frac{1}{5} & -\frac{1}{5} & -\frac{3}{20} \\ \frac{1}{5} & -\frac{1}{5} & -\frac{2}{5} \end{pmatrix}.$$

✓ *Check:* Row 3 of $A-B$ times column 3: $0(\frac{7}{40}) + 4(-\frac{3}{20}) + (-4)(-\frac{2}{5}) = -0.6 + 1.6 = 1$.

Answer. (a) $A^{-1} = \frac{1}{30} \begin{pmatrix} 3 & 1 & -11 \\ 6 & 2 & 8 \\ -9 & 7 & 13 \end{pmatrix}$, $B^{-1} = \frac{1}{5} \begin{pmatrix} 23 & 8 & -2 \\ -14 & -4 & 1 \\ 1 & 1 & 1 \end{pmatrix}$, and $(AB)^{-1} = B^{-1}A^{-1}$ as displayed. (b) No. (c) Yes, always. (d) Yes, $\det(A+B) = -60$. (e) Yes, $\det(A-B) = 40$.

Where marks are lost

The adjoint is the **transpose** of the cofactor matrix. Forgetting the transpose produces a matrix that fails $AA^{-1} = I$ in exactly the off-diagonal entries — which is why the check above tests an off-diagonal entry, not just a diagonal one.

6.3 Four inverses by cofactors**Q.3**

Find the inverse, if it exists, of the given matrices using both the cofactor approach and Gaussian reduction.

$$(a) \begin{pmatrix} 6 & 0 & -1 \\ 1 & 3 & 5 \\ -3 & 1 & 0 \end{pmatrix} \quad (b) \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{pmatrix} \quad (c) \begin{pmatrix} -1 & 2 & -3 \\ 0 & 1 & 5 \\ 2 & 4 & 7 \end{pmatrix} \quad (d) \begin{pmatrix} 8 & 0 & -2 \\ 1 & -2 & 1 \\ -3 & 1 & 1 \end{pmatrix}$$

Solution

The cofactor route is shown in full for (a) and summarised for the rest; the Gaussian route — row-reducing $[A \mid I]$ to $[I \mid A^{-1}]$ — gives identical answers and is sketched at the end.

(a) $\det = 6(0 - 5) - 0 + (-1)(1 + 9) = -30 - 10 = -40$. Cofactors:

$$\begin{aligned} C_{11} &= -5, & C_{12} &= -15, & C_{13} &= 10, \\ C_{21} &= -1, & C_{22} &= -3, & C_{23} &= -6, \\ C_{31} &= 3, & C_{32} &= -31, & C_{33} &= 18. \end{aligned}$$

Transposing and dividing by $\det$:

$$A^{-1} = \frac{1}{-40} \begin{pmatrix} -5 & -1 & 3 \\ -15 & -3 & -31 \\ 10 & -6 & 18 \end{pmatrix} = \frac{1}{40} \begin{pmatrix} 5 & 1 & -3 \\ 15 & 3 & 31 \\ -10 & 6 & -18 \end{pmatrix}.$$

✓ *Check:* Row 1 $(6, 0, -1)$ times column 1 $\frac{1}{40}(5, 15, -10)^t$: $\frac{30+0+10}{40} = 1$. Row 2 $(1, 3, 5)$ times the same column: $\frac{5+45-50}{40} = 0$.

(b) $\det = 1(36 - 25) - 2(12 - 5) + 3(5 - 3) = 11 - 14 + 6 = 3$.

$$A^{-1} = \frac{1}{3} \begin{pmatrix} 11 & -9 & 1 \\ -7 & 9 & -2 \\ 2 & -3 & 1 \end{pmatrix}.$$

✓ *Check:* Row 3 $(1, 5, 12)$ times column 1 $\frac{1}{3}(11, -7, 2)^t$: $\frac{11-35+24}{3} = 0$. Row 1 times column 1: $\frac{11-14+6}{3} = 1$.

(c) $\det = -1(7 - 20) - 2(0 - 10) + (-3)(0 - 2) = 13 + 20 + 6 = 39$.

$$A^{-1} = \frac{1}{39} \begin{pmatrix} -13 & -26 & 13 \\ 10 & -1 & 5 \\ -2 & 8 & -1 \end{pmatrix}.$$

✓ *Check:* Row 1 $(-1, 2, -3)$ times column 1 $\frac{1}{39}(-13, 10, -2)^t$: $\frac{13+20+6}{39} = 1$. Row 2 $(0, 1, 5)$ times the same: $\frac{0+10-10}{39} = 0$.

(d) $\det = 8(-2 - 1) - 0 + (-2)(1 - 6) = -24 + 10 = -14$.

$$A^{-1} = \frac{1}{-14} \begin{pmatrix} -3 & -2 & -4 \\ -4 & 2 & -10 \\ -5 & -8 & -16 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 3 & 2 & 4 \\ 4 & -2 & 10 \\ 5 & 8 & 16 \end{pmatrix}.$$

✓ *Check:* Row 1 $(8, 0, -2)$ times column 1 $\frac{1}{14}(3, 4, 5)^t$: $\frac{24+0-10}{14} = 1$. Row 3 $(-3, 1, 1)$ times the same: $\frac{-9+4+5}{14} = 0$.

The Gaussian route, on (b). Augment with the identity and reduce:

$$\begin{aligned} & \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 1 & 3 & 5 & 0 & 1 & 0 \\ 1 & 5 & 12 & 0 & 0 & 1 \end{array} \right] \xrightarrow[R_3-R_1]{R_2-R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 3 & 9 & -1 & 0 & 1 \end{array} \right] \\ & \xrightarrow{R_3-3R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 3 & 2 & -3 & 1 \end{array} \right] \xrightarrow{\frac{1}{3}R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 1 & \frac{2}{3} & -1 & \frac{1}{3} \end{array} \right], \end{aligned}$$

and back-substituting ($R_2 - 2R_3$, then $R_1 - 3R_3$, then $R_1 - 2R_2$) yields

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{11}{3} & -3 & \frac{1}{3} \\ 0 & 1 & 0 & -\frac{7}{3} & 3 & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{2}{3} & -1 & \frac{1}{3} \end{array} \right],$$

which is $\frac{1}{3} \begin{pmatrix} 11 & -9 & 1 \\ -7 & 9 & -2 \\ 2 & -3 & 1 \end{pmatrix}$ — the cofactor answer.

Answer. (a) $\frac{1}{40} \begin{pmatrix} 5 & 1 & -3 \\ 15 & 3 & 31 \\ -10 & 6 & -18 \end{pmatrix}$ (b) $\frac{1}{3} \begin{pmatrix} 11 & -9 & 1 \\ -7 & 9 & -2 \\ 2 & -3 & 1 \end{pmatrix}$ (c) $\frac{1}{39} \begin{pmatrix} -13 & -26 & 13 \\ 10 & -1 & 5 \\ -2 & 8 & -1 \end{pmatrix}$
 (d) $\frac{1}{14} \begin{pmatrix} 3 & 2 & 4 \\ 4 & -2 & 10 \\ 5 & 8 & 16 \end{pmatrix}$. All four determinants are non-zero, so all four inverses exist.

6.4 Systems by matrices and by Cramer's rule

Q.4

Solve the following systems of linear equations using matrices as well as Cramer's rule.

$$\begin{array}{lll}
 2x - y + z = 5 & x + y = 2 & 2x + y + 3z = 3 \\
 \text{(a) } 4x + 2y + 3z = 8 & \text{(b) } 2x - z = 1 & \text{(c) } x + y - 2z = 0 \\
 3x - 4y - z = 3 & 2y - 3z = -1 & -3x - y + 2z = -4
 \end{array}$$

Solution

(a) The coefficient determinant, expanded along row 1:

$$D = \begin{vmatrix} 2 & -1 & 1 \\ 4 & 2 & 3 \\ 3 & -4 & -1 \end{vmatrix} = 2(-2 + 12) + 1(-4 - 9) + 1(-16 - 6) = 20 - 13 - 22 = -15.$$

$D \neq 0$, so Cramer's rule applies. Replace one column at a time by the right-hand side $(5, 8, 3)^t$:

$$D_x = \begin{vmatrix} 5 & -1 & 1 \\ 8 & 2 & 3 \\ 3 & -4 & -1 \end{vmatrix} = 5(10) + 1(-17) + 1(-38) = -5,$$

$$D_y = \begin{vmatrix} 2 & 5 & 1 \\ 4 & 8 & 3 \\ 3 & 3 & -1 \end{vmatrix} = 2(-17) - 5(-13) + 1(-12) = 19,$$

$$D_z = \begin{vmatrix} 2 & -1 & 5 \\ 4 & 2 & 8 \\ 3 & -4 & 3 \end{vmatrix} = 2(38) + 1(-12) + 5(-22) = -46.$$

$$x = \frac{D_x}{D} = \frac{-5}{-15} = \frac{1}{3}, \quad y = \frac{D_y}{D} = \frac{19}{-15} = -\frac{19}{15}, \quad z = \frac{D_z}{D} = \frac{-46}{-15} = \frac{46}{15}.$$

✓ *Check:* Equation 1: $2(\frac{1}{3}) + \frac{19}{15} + \frac{46}{15} = \frac{10+19+46}{15} = \frac{75}{15} = 5$. Equation 3: $1 + \frac{76}{15} - \frac{46}{15} = 1 + 2 = 3$. Both hold.

(b) Write the missing variables in explicitly — the zeros are what make this system easy:

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & -1 \\ 0 & 2 & -3 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}.$$

$$D = 1(0 + 2) - 1(-6 - 0) + 0 = 2 + 6 = 8.$$

$$D_x = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 0 & -1 \\ -1 & 2 & -3 \end{vmatrix} = 8, \quad D_y = \begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & -1 \\ 0 & -1 & -3 \end{vmatrix} = 8, \quad D_z = \begin{vmatrix} 1 & 1 & 2 \\ 2 & 0 & 1 \\ 0 & 2 & -1 \end{vmatrix} = 8.$$

$$x = \frac{8}{8} = 1, \quad y = \frac{8}{8} = 1, \quad z = \frac{8}{8} = 1.$$

✓ *Check:* $1 + 1 = 2$; $2(1) - 1 = 1$; $2(1) - 3(1) = -1$. All three hold.

(c)

$$D = \begin{vmatrix} 2 & 1 & 3 \\ 1 & 1 & -2 \\ -3 & -1 & 2 \end{vmatrix} = 2(2 - 2) - 1(2 - 6) + 3(-1 + 3) = 0 + 4 + 6 = 10.$$

$$D_x = \begin{vmatrix} 3 & 1 & 3 \\ 0 & 1 & -2 \\ -4 & -1 & 2 \end{vmatrix} = 20, \quad D_y = \begin{vmatrix} 2 & 3 & 3 \\ 1 & 0 & -2 \\ -3 & -4 & 2 \end{vmatrix} = -16, \quad D_z = \begin{vmatrix} 2 & 1 & 3 \\ 1 & 1 & 0 \\ -3 & -1 & -4 \end{vmatrix} = 2.$$

$$x = \frac{20}{10} = 2, \quad y = \frac{-16}{10} = -\frac{8}{5}, \quad z = \frac{2}{10} = \frac{1}{5}.$$

✓ *Check:* Equation 1: $4 - \frac{8}{5} + \frac{3}{5} = 4 - 1 = 3$. Equation 2: $2 - \frac{8}{5} - \frac{2}{5} = 2 - 2 = 0$. Equation 3: $-6 + \frac{8}{5} + \frac{2}{5} = -6 + 2 = -4$. All three hold.

The matrix method gives the same thing. $Ax = b \Rightarrow x = A^{-1}b$. For (c), $\det A = 10$ and $A^{-1} = \frac{1}{10} \text{adj } A$; multiplying by $b = (3, 0, -4)^t$ reproduces $(2, -\frac{8}{5}, \frac{1}{5})^t$. Cramer's rule is simply this computation with the common denominator $\det A$ kept visible, which is why it is faster when only one variable is wanted.

Answer. (a) $x = \frac{1}{3}, y = -\frac{19}{15}, z = \frac{46}{15}$ (b) $x = y = z = 1$ (c) $x = 2, y = -\frac{8}{5}, z = \frac{1}{5}$

Where marks are lost

In system (b), the equations are written with variables missing. Insert the zero coefficients ($x + y + 0z = 2, 2x + 0y - z = 1, 0x + 2y - 3z = -1$) before forming D . Reading the coefficients straight off the printed equations puts numbers in the wrong columns and is the commonest way to lose all the marks on an otherwise easy question.

6.5 An electronics production plan — and a defective question

Q.5

An electronic company manufactures resistors, transistors and computer chips. Each resistor requires 3 units of copper, 2 units of zinc and 1 unit of glass. Each transistor requires 3, 2 and 1 units of each material respectively, and each computer chip requires 2, 1 and 2 units of these materials respectively. Using 800 units of copper, 400 units of zinc and 500 units of glass, how many of each item can be prepared?

Solution

Let r, t and c be the numbers of resistors, transistors and chips. Reading the requirements off the question:

$$\begin{array}{ll} \text{copper :} & 3r + 3t + 2c = 800 \\ \text{zinc :} & 2r + 2t + c = 400 \\ \text{glass :} & r + t + 2c = 500 \end{array} \quad A = \begin{pmatrix} 3 & 3 & 2 \\ 2 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

The coefficient matrix is singular. Columns 1 and 2 of A are identical, because the question gives resistors and transistors the *same* requirement $(3, 2, 1)$. Therefore

$$\det A = 0,$$

and Cramer's rule cannot be applied at all. Structurally, the two products are mathematically indistinguishable: no amount of material data can tell a resistor from a transistor, so r and t can never be determined separately — only their sum.

And the system is inconsistent. Put $s = r + t$. The three equations become

$$3s + 2c = 800, \quad 2s + c = 400, \quad s + 2c = 500.$$

From the second, $c = 400 - 2s$. Substituting into the first,

$$3s + 2(400 - 2s) = 800 \implies 3s + 800 - 4s = 800 \implies -s = 0 \implies s = 0,$$

so $c = 400$. Testing these in the third equation:

$$s + 2c = 0 + 800 = 800 \neq 500.$$

The third equation is violated, so no (r, t, c) satisfies all three.

Answer. As printed, the question has **no solution**. The coefficient matrix is singular ($\det A = 0$, columns 1 and 2 identical, because resistors and transistors are given identical material requirements), and the reduced system in $s = r + t$ and c is inconsistent: it forces $s = 0$, $c = 400$, which fails the glass constraint 500.

The repaired problem. The exercise is clearly intended to have distinct requirements, and the smallest edit that achieves it is to change the transistor's zinc requirement from 2 to 1 — so a transistor needs 3 copper, 1 zinc, 1 glass. The system becomes

$$\begin{aligned} 3r + 3t + 2c &= 800 \\ 2r + t + c &= 400 \\ r + t + 2c &= 500 \end{aligned} \quad A = \begin{pmatrix} 3 & 3 & 2 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix},$$

$$D = 3(2 - 1) - 3(4 - 1) + 2(2 - 1) = 3 - 9 + 2 = -4 \neq 0,$$

so Cramer's rule now runs normally:

$$D_r = \begin{vmatrix} 800 & 3 & 2 \\ 400 & 1 & 1 \\ 500 & 1 & 2 \end{vmatrix} = -300, \quad D_t = \begin{vmatrix} 3 & 800 & 2 \\ 2 & 400 & 1 \\ 1 & 500 & 2 \end{vmatrix} = -300, \quad D_c = \begin{vmatrix} 3 & 3 & 800 \\ 2 & 1 & 400 \\ 1 & 1 & 500 \end{vmatrix} = -700,$$

$$r = \frac{-300}{-4} = 75, \quad t = \frac{-300}{-4} = 75, \quad c = \frac{-700}{-4} = 175.$$

✓ *Check:* Copper: $3(75) + 3(75) + 2(175) = 225 + 225 + 350 = 800$. Zinc: $2(75) + 75 + 175 = 400$. Glass: $75 + 75 + 2(175) = 500$. All three materials are consumed exactly.

The *method* — form A , test $\det A$, apply Cramer — is the examinable content, and it is unaffected. The printed data simply is not solvable.

Where marks are lost

Compute D *first*, always. If $D = 0$, Cramer's rule is unavailable and the correct answer is a statement about the system — “no solution” or “infinitely many” — not a set of

numbers. Producing numbers from a singular system means an arithmetic error has been made somewhere.

6.6 Machine time and material cost

Q.6

In a pottery factory, a machine takes 3 minutes to make a glass and 2 minutes to make a plate. The material for a glass costs \$0.25 and for a plate \$0.20. If the machine runs for 10 hours and exactly \$50 is used on material, how many glasses and plates can be produced?

Solution

Let g be the number of glasses and p the number of plates. Ten hours is 600 minutes, so:

$$\begin{array}{ll} \text{time :} & 3g + 2p = 600 \\ \text{material :} & 0.25g + 0.20p = 50 \end{array}$$

Clear the decimals in the second equation by multiplying by 20:

$$5g + 4p = 1000.$$

By elimination. Double the time equation so the p terms match:

$$6g + 4p = 1200.$$

Subtracting the material equation:

$$(6g + 4p) - (5g + 4p) = 1200 - 1000 \implies g = 200.$$

Back into $3g + 2p = 600$:

$$600 + 2p = 600 \implies p = 0.$$

✓ *Check:* Time: $3(200) + 2(0) = 600$ minutes = 10 hours. Material: $0.25(200) + 0.20(0) = \$50$. Both constraints are met exactly.

Reading the answer. The solution sits on the boundary of the feasible region: the two constraints are satisfied only by making glasses alone. The system is not degenerate — $D = \begin{vmatrix} 3 & 2 \\ 5 & 4 \end{vmatrix} = 12 - 10 = 2 \neq 0$, so the solution is unique — it simply happens that the two constraints bind simultaneously at a corner where $p = 0$.

Answer. $g = 200$ glasses and $p = 0$ plates. Running for 10 hours on exactly \$50 of material is only possible by producing 200 glasses and no plates.

Limits and Differentiation

Source: *Business Mathematics* 5405/1429, §7.14 SELF ASSESSMENT QUESTIONS, pp. 167–169. Several sub-parts of Q.6 are illegible in the distributed PDF (radicals and fraction bars collapsed in the text layer); those are named and skipped rather than guessed at.

The three limit situations, and what to do in each

Substitution works If f is continuous at a , then $\lim_{x \rightarrow a} f(x) = f(a)$. Try this first, every time.

You get $\frac{0}{0}$ The zero factors cancel. Factorise, or rationalise a surd, then substitute again.

$x \rightarrow \infty$ Divide numerator and denominator by the highest power of x in the *denominator*, then use $\frac{1}{x^n} \rightarrow 0$.

And the derivative rules the unit tests:

$$\frac{d}{dx} x^n = nx^{n-1}, \quad (uv)' = u'v + uv', \quad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}, \quad \frac{d}{dx} f(g(x)) = f'(g(x)) g'(x).$$

7.1 Estimating limits from a table

Q.1

Estimate the following limits by constructing a table. (a) $\lim_{x \rightarrow 2} (x - 3)$ (b) $\lim_{x \rightarrow 1} \frac{x}{\ln x - 1}$ (c)

$$\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2}$$

Solution

A table approaches a from both sides. If the two columns settle on the same number, that number is the limit.

(a) $f(x) = x - 3$ is a polynomial, hence continuous everywhere, so substitution is legitimate: $\lim_{x \rightarrow 2} (x - 3) = 2 - 3 = -1$.

$x \rightarrow 2^-$	$f(x)$	$x \rightarrow 2^+$	$f(x)$
1.9	-1.1	2.1	-0.9
1.99	-1.01	2.01	-0.99
1.999	-1.001	2.001	-0.999

Both columns close on -1 .

(b) At $x = 1$, $\ln 1 = 0$, so the denominator is $0 - 1 = -1$, not zero. There is no indeterminacy at all and substitution applies:

$$\lim_{x \rightarrow 1} \frac{x}{\ln x - 1} = \frac{1}{0 - 1} = -1.$$

$x \rightarrow 1^-$	$f(x)$	$x \rightarrow 1^+$	$f(x)$
0.9	-0.8142	1.1	-1.2159
0.99	-0.9802	1.01	-1.0202
0.999	-0.9980	1.001	-1.0020

(c) Substituting $x = -2$ gives $\frac{4-4}{0} = \frac{0}{0}$ — indeterminate. Factorise and cancel:

$$\frac{x^2 - 4}{x + 2} = \frac{(x - 2)(x + 2)}{x + 2} = x - 2 \quad (x \neq -2),$$

$$\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2} = \lim_{x \rightarrow -2} (x - 2) = -4.$$

$x \rightarrow -2^-$	$f(x)$	$x \rightarrow -2^+$	$f(x)$
-2.1	-4.1	-1.9	-3.9
-2.01	-4.01	-1.99	-3.99
-2.001	-4.001	-1.999	-3.999

Answer. (a) -1 (b) -1 (c) -4

Where marks are lost

In (c) the function is *undefined* at $x = -2$ — the cancellation is valid only for $x \neq -2$. A limit describes the approach, never the value at the point, and the two differ here.

7.2 Evaluating limits algebraically

Q.2

Find the indicated limits. (a) $\lim_{x \rightarrow 3} \frac{x^4 - 81}{x^2 - x - 6}$ (b) $\lim_{x \rightarrow \infty} \frac{4 - 3x^3}{x^3 - 1}$ (c) $\lim_{x \rightarrow -7^-} \frac{x^2 + 1}{x^2 - 49}$

(e) Find the value of the constant k such that $\lim_{x \rightarrow 2} f(x)$ exists, if $f(x) = \begin{cases} 2 - x, & x < 2 \\ x^3 + k(x + 1), & x \geq 2 \end{cases}$

(f) Use the graph of f to find each of the following limits if they exist: (i) $\lim_{x \rightarrow -1} f(x)$, (ii) $\lim_{x \rightarrow 0} f(x)$, (iii) $\lim_{x \rightarrow 1} f(x)$, where $f(x) = \begin{cases} x^2 + 1, & x < -1 \\ 3x + 2, & -1 \leq x \leq 1 \\ 2 - x, & x > 1 \end{cases}$

Part (d) is illegible in the distributed PDF and is omitted.

Solution

(a) Substitution gives $\frac{81-81}{9-3-6} = \frac{0}{0}$. Factorise both parts, using the difference of two squares twice on top:

$$x^4 - 81 = (x^2 - 9)(x^2 + 9) = (x - 3)(x + 3)(x^2 + 9), \quad x^2 - x - 6 = (x - 3)(x + 2).$$

The common factor $(x - 3)$ cancels:

$$\lim_{x \rightarrow 3} \frac{(x-3)(x+3)(x^2+9)}{(x-3)(x+2)} = \lim_{x \rightarrow 3} \frac{(x+3)(x^2+9)}{x+2} = \frac{(6)(18)}{5} = \frac{108}{5} = 21.6.$$

(b) As $x \rightarrow \infty$ this is $\frac{\infty}{\infty}$. Divide top and bottom by x^3 :

$$\lim_{x \rightarrow \infty} \frac{4 - 3x^3}{x^3 - 1} = \lim_{x \rightarrow \infty} \frac{\frac{4}{x^3} - 3}{1 - \frac{1}{x^3}} = \frac{0 - 3}{1 - 0} = -3.$$

Equivalently: for a ratio of polynomials of equal degree, the limit at infinity is the ratio of leading coefficients, $\frac{-3}{1}$.

(c) A one-sided limit, approaching -7 from the *left* ($x < -7$). The numerator tends to $49 + 1 = 50$, a positive number. For the denominator, $x < -7$ means $|x| > 7$, so $x^2 > 49$ and

$$x^2 - 49 \rightarrow 0^+.$$

A fixed positive number divided by something positive and shrinking to zero grows without bound:

$$\lim_{x \rightarrow -7^-} \frac{x^2 + 1}{x^2 - 49} = +\infty.$$

✓ *Check:* At $x = -7.01$: $x^2 - 49 = 0.1401$ and $x^2 + 1 = 50.14$, giving 357.9. At $x = -7.001$: the quotient is 3572. Growing, and positive.

(e) A limit exists at $x = 2$ exactly when the two one-sided limits agree.

$$\lim_{x \rightarrow 2^-} f(x) = 2 - 2 = 0, \quad \lim_{x \rightarrow 2^+} f(x) = 2^3 + k(2 + 1) = 8 + 3k.$$

Setting them equal:

$$8 + 3k = 0 \implies k = -\frac{8}{3}.$$

✓ *Check:* With $k = -\frac{8}{3}$: $f(2^+) = 8 - \frac{8}{3}(3) = 8 - 8 = 0 = f(2^-)$. The limit exists and equals 0 — and since $f(2) = 0$ too, f is in fact continuous there.

(f) Check the two one-sided limits at each junction of the piecewise definition.

	Left-hand limit	Right-hand limit	Conclusion
(i) $x \rightarrow -1$	$(-1)^2 + 1 = 2$	$3(-1) + 2 = -1$	unequal $\Rightarrow$ limit does not exist
(ii) $x \rightarrow 0$	$3(0) + 2 = 2$	$3(0) + 2 = 2$	both from $3x + 2 \Rightarrow$ limit = 2
(iii) $x \rightarrow 1$	$3(1) + 2 = 5$	$2 - 1 = 1$	unequal $\Rightarrow$ limit does not exist

At $x = 0$ the point lies strictly inside the middle piece, so there is no junction and the limit is just the value: 2.

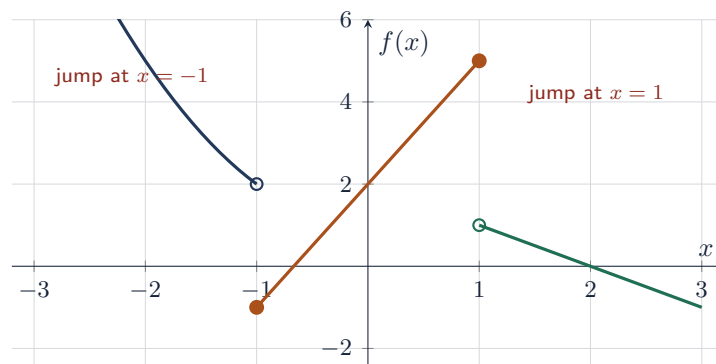


Figure 7.1: The piecewise function of Q.2(f). The limit fails to exist at $x = -1$ and $x = 1$ because the graph jumps; at $x = 0$ it is continuous.

Answer. (a) $\frac{108}{5} = 21.6$ (b) -3 (c) $+\infty$ (e) $k = -\frac{8}{3}$ (f) (i) does not exist, (ii) 2, (iii) does not exist.

Where marks are lost

In (c) the side matters. From the *right* ($-7 < x < 7$) we would have $x^2 < 49$, so $x^2 - 49 \rightarrow 0^-$ and the limit would be $-\infty$. A one-sided limit question is asking you to determine a sign, and that is where the mark is.

7.3 A population in the long run

Q.3

The population of a certain village t years from now is expected to be

$$N = 15000 + \frac{5000}{(t+3)^2}.$$

Determine $\lim_{t \rightarrow \infty} N$, i.e. the population in the long run.

Solution

Only the second term depends on t . As t grows, $(t+3)^2$ grows without bound, so the fraction — a fixed numerator over an exploding denominator — shrinks to zero:

$$\lim_{t \rightarrow \infty} \frac{5000}{(t+3)^2} = 0.$$

$$\lim_{t \rightarrow \infty} N = 15000 + 0 = 15000.$$

Interpretation. The population declines towards a floor of 15,000 and never falls below it. Today ($t = 0$) it is $15000 + \frac{5000}{9} \approx 15,556$; after 10 years, $15000 + \frac{5000}{169} \approx 15,030$. The line $N = 15000$ is a horizontal asymptote approached from above.

Answer. $\lim_{t \rightarrow \infty} N = 15,000$. The population settles at 15,000 in the long run, approaching that level from above.

7.4 Average cost per item

Q.4

The cost of manufacturing a certain item is $C(x) = 10000 + 6x$, where x is the number of items produced. The average cost per item, $\bar{C}(x)$, is obtained by dividing $C(x)$ by x . Determine (a) $\bar{C}(1500)$, (b) $\bar{C}(10,000)$, (c) $\lim_{x \rightarrow 5000} \bar{C}(x)$.

Solution

$$\bar{C}(x) = \frac{C(x)}{x} = \frac{10000 + 6x}{x} = \frac{10000}{x} + 6.$$

Written this way the structure is plain: a fixed cost of \$10,000 spread over x items, plus \$6 of variable cost that never spreads.

$$(a) \bar{C}(1500) = \frac{10000}{1500} + 6 = 6.667 + 6 = \$12.67.$$

$$(b) \bar{C}(10000) = \frac{10000}{10000} + 6 = 1 + 6 = \$7.00.$$

(c) $\bar{C}$ is continuous at $x = 5000$ (the only discontinuity is at $x = 0$), so the limit is the value:

$$\lim_{x \rightarrow 5000} \bar{C}(x) = \frac{10000}{5000} + 6 = 2 + 6 = \$8.00.$$

Worth adding. As $x \rightarrow \infty$, $\bar{C}(x) \rightarrow 6$: average cost falls towards the variable cost per unit but never reaches it. That is the whole economics of spreading a fixed cost, and it is a standard follow-up part.

Answer. (a) \$12.67 (b) \$7.00 (c) \$8.00; and $\bar{C}(x) \rightarrow \$6$ as $x \rightarrow \infty$.

7.5 Average versus instantaneous rate of change

Q.5

The cost of manufacturing x units of a certain item is $C(x) = x^2 - 5x + 80$. (a) Find the average rate of change with respect to the number of items produced between 10 and 15. (b) Find the instantaneous rate of change when 12 items are produced.

Solution

(a) **Average rate of change** is the slope of the chord joining the two points — total change divided by total change in x :

$$C(15) = 225 - 75 + 80 = 230, \quad C(10) = 100 - 50 + 80 = 130.$$

$$\frac{\Delta C}{\Delta x} = \frac{C(15) - C(10)}{15 - 10} = \frac{230 - 130}{5} = \frac{100}{5} = 20.$$

Cost rises by \$20 per additional unit, *averaged* over that range.

(b) **Instantaneous rate of change** is the derivative — the slope of the tangent at a single point. This is the *marginal cost*:

$$C'(x) = 2x - 5, \quad C'(12) = 24 - 5 = 19.$$

The 13th unit costs about \$19 to produce.

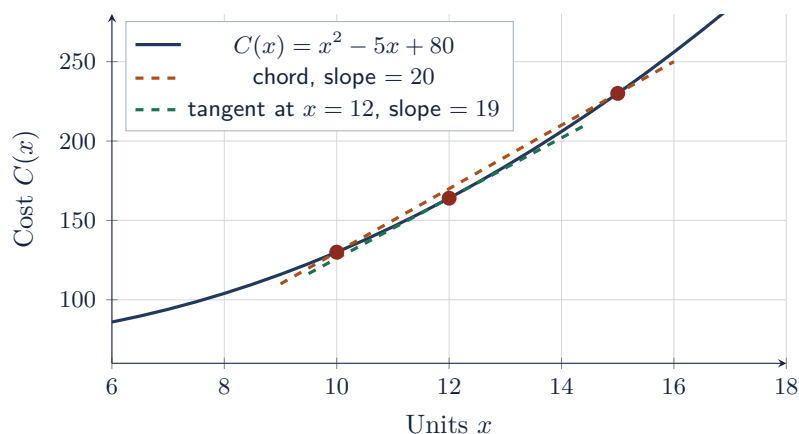


Figure 7.2: The chord from $x = 10$ to $x = 15$ has slope 20 (average rate); the tangent at $x = 12$ has slope 19 (instantaneous rate).

Answer. (a) 20 per unit (b) $C'(12) = 19$ per unit

Where marks are lost

“Average rate of change” needs two points and a subtraction; “instantaneous rate of change” needs one point and a derivative. They are different quantities and give different numbers (20 versus 19 here). Differentiating in part (a), or substituting into C in part (b), answers the wrong question.

7.6 Differentiation

Q.6 (c)

Find the derivative of $f(x) = \sqrt{x^3 + 2x^2 + 3}$.

Only sub-part (c) of Q.6 survives legibly in the distributed PDF. Sub-parts (a), (b), (d), (e) and (f) have collapsed radicals and fraction bars in the text layer and cannot be transcribed with confidence, so they are omitted. The drill below exercises the same four rules on clean functions.

Solution

This is a chain rule problem: an outer square root wrapped around an inner polynomial. Write $f(x) = u^{1/2}$ with $u = x^3 + 2x^2 + 3$.

$$\frac{df}{dx} = \frac{1}{2}u^{-1/2} \cdot \frac{du}{dx} = \frac{1}{2\sqrt{u}} (3x^2 + 4x),$$

$$f'(x) = \frac{3x^2 + 4x}{2\sqrt{x^3 + 2x^2 + 3}}.$$

Answer. $f'(x) = \frac{3x^2 + 4x}{2\sqrt{x^3 + 2x^2 + 3}}$

Drill on the same four rules

The following are worked in the style of Q.6 and cover exactly the rules the unit examines. They are set here, not taken from the book.

Power and sum rules

$$y = 4x^5 - \frac{7}{x^2} + 3\sqrt{x} - 9 = 4x^5 - 7x^{-2} + 3x^{1/2} - 9.$$

Rewriting every term as a power of x *before* differentiating is the whole technique:

$$\frac{dy}{dx} = 20x^4 + 14x^{-3} + \frac{3}{2}x^{-1/2} = 20x^4 + \frac{14}{x^3} + \frac{3}{2\sqrt{x}}.$$

Product rule

$$y = (5x - 2)(3x^2 + 2).$$

With $u = 5x - 2$, $u' = 5$ and $v = 3x^2 + 2$, $v' = 6x$:

$$\frac{dy}{dx} = u'v + uv' = 5(3x^2 + 2) + (5x - 2)(6x) = 15x^2 + 10 + 30x^2 - 12x = 45x^2 - 12x + 10.$$

✓ *Check:* Expanding first, $y = 15x^3 - 6x^2 + 10x - 4$, so $y' = 45x^2 - 12x + 10$. The two routes agree.

Quotient rule

$$y = \frac{8x^2 - 5}{x^2 - 3}.$$

With $u = 8x^2 - 5$, $u' = 16x$ and $v = x^2 - 3$, $v' = 2x$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{u'v - uv'}{v^2} = \frac{16x(x^2 - 3) - (8x^2 - 5)(2x)}{(x^2 - 3)^2} \\ &= \frac{16x^3 - 48x - 16x^3 + 10x}{(x^2 - 3)^2} = \frac{-38x}{(x^2 - 3)^2}. \end{aligned}$$

Chain rule inside a product

$$y = (6x^3 - x^2 + 3x + 5)^2.$$

Outer function squared, inner polynomial differentiated:

$$\frac{dy}{dx} = 2(6x^3 - x^2 + 3x + 5)^1 \cdot (18x^2 - 2x + 3) = 2(6x^3 - x^2 + 3x + 5)(18x^2 - 2x + 3).$$

Where marks are lost

The quotient rule's numerator is $u'v - uv'$, and the order matters — swapping the two terms flips the sign of the whole derivative. A useful memory hook: the derivative of the *top* comes first.

Partial Derivatives

Source: *Business Mathematics* 5405/1429, §8.9 SELF ASSESSMENT QUESTIONS, pp. 183–184.

Hold the other variable constant

To find f_x , differentiate with respect to x and treat y as a *number*. To find f_y , treat x as a number. Every ordinary rule — power, product, quotient, chain — carries over unchanged.

For mixed partials, **Clairaut's theorem** says the order of differentiation does not matter for the functions in this course:

$$f_{xy} = f_{yx}, \quad f_{xyz} = f_{yzx} = f_{xzy} = \dots$$

That is a free correctness check, and Question 3 is built on it.

Second-derivative test for $z = f(x, y)$ at a critical point where $f_x = f_y = 0$: with $D = f_{xx}f_{yy} - (f_{xy})^2$,

$D > 0$ and $f_{xx} > 0$
relative minimum

$D > 0$ and $f_{xx} < 0$
relative maximum

$D < 0$
saddle point

and $D = 0$ is inconclusive — the test simply cannot decide.

8.1 First partial derivatives

Q.1

Find f_x , f_y , $f_x(3, 4)$ and $f_y(1, 2)$ for (a) $f(x, y) = \frac{x + 3xy^2}{x^3 + y^2}$ (b) $f(x, y) = y e^{5x+2y^2}$ (c) $f(x, y) = (x + 1)^2 + (y - 3)^3 + 5xy^3 - 2$

Solution

(a) Quotient rule, with y held constant for f_x .

$$u = x + 3xy^2 = x(1 + 3y^2), \quad u_x = 1 + 3y^2; \quad v = x^3 + y^2, \quad v_x = 3x^2.$$

$$f_x = \frac{(1 + 3y^2)(x^3 + y^2) - x(1 + 3y^2)(3x^2)}{(x^3 + y^2)^2}.$$

The factor $(1 + 3y^2)$ is common to both terms of the numerator — taking it out collapses the whole expression:

$$f_x = \frac{(1 + 3y^2)[(x^3 + y^2) - 3x^3]}{(x^3 + y^2)^2} = \frac{(1 + 3y^2)(y^2 - 2x^3)}{(x^3 + y^2)^2}.$$

$$f_x(3, 4) = \frac{(1 + 48)(16 - 54)}{(27 + 16)^2} = \frac{49(-38)}{1849} = -\frac{1862}{1849} \approx -1.0070.$$

Now f_y , with x held constant. Here $u_y = 6xy$ and $v_y = 2y$:

$$f_y = \frac{6xy(x^3 + y^2) - x(1 + 3y^2)(2y)}{(x^3 + y^2)^2} = \frac{2xy[3(x^3 + y^2) - (1 + 3y^2)]}{(x^3 + y^2)^2},$$

and the $3y^2$ terms cancel inside the bracket:

$$f_y = \frac{2xy(3x^3 - 1)}{(x^3 + y^2)^2}.$$

$$f_y(1, 2) = \frac{2(1)(2)(3 - 1)}{(1 + 4)^2} = \frac{8}{25} = 0.32.$$

(b) $f = ye^{5x+2y^2}$.

For f_x , the factor y is a constant and the chain rule gives $\frac{\partial}{\partial x}(5x + 2y^2) = 5$:

$$f_x = 5ye^{5x+2y^2}.$$

For f_y , both the leading y and the exponent depend on y , so the product rule is needed, with $\frac{\partial}{\partial y}(5x + 2y^2) = 4y$:

$$f_y = (1)e^{5x+2y^2} + y(4y)e^{5x+2y^2} = (1 + 4y^2)e^{5x+2y^2}.$$

$$f_x(3, 4) = 5(4)e^{15+32} = 20e^{47}, \quad f_y(1, 2) = (1 + 16)e^{5+8} = 17e^{13} \approx 7.55 \times 10^6.$$

(c) $f = (x + 1)^2 + (y - 3)^3 + 5xy^3 - 2$. Term by term:

$$f_x = 2(x + 1) + 5y^3, \quad f_y = 3(y - 3)^2 + 15xy^2.$$

The term $(y - 3)^3$ vanishes under $\partial/\partial x$ because it contains no x ; likewise $(x + 1)^2$ vanishes under $\partial/\partial y$.

$$f_x(3, 4) = 2(4) + 5(64) = 8 + 320 = 328,$$

$$f_y(1, 2) = 3(2 - 3)^2 + 15(1)(4) = 3 + 60 = 63.$$

Answer. (a) $f_x = \frac{(1 + 3y^2)(y^2 - 2x^3)}{(x^3 + y^2)^2}$, $f_y = \frac{2xy(3x^3 - 1)}{(x^3 + y^2)^2}$; $f_x(3, 4) = -\frac{1862}{1849} \approx -1.007$,

$$f_y(1, 2) = \frac{8}{25} = 0.32.$$

(b) $f_x = 5ye^{5x+2y^2}$, $f_y = (1 + 4y^2)e^{5x+2y^2}$; $f_x(3, 4) = 20e^{47}$, $f_y(1, 2) = 17e^{13}$.

(c) $f_x = 2(x + 1) + 5y^3$, $f_y = 3(y - 3)^2 + 15xy^2$; $f_x(3, 4) = 328$, $f_y(1, 2) = 63$.

Where marks are lost

In (b), f_y needs the product rule because y appears twice — once as a coefficient and once inside the exponent. Writing $f_y = 4y^2e^{5x+2y^2}$ drops the term from differentiating the leading y .

8.2 Verifying Laplace's equation

Q.2

If $z = x^2 - y^2 + 2xy$, show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$.

Solution

Differentiate twice with respect to each variable in turn.

With respect to x (treating y as constant, so $-y^2$ vanishes and $2xy$ gives $2y$):

$$\frac{\partial z}{\partial x} = 2x + 2y \implies \frac{\partial^2 z}{\partial x^2} = 2.$$

With respect to y (treating x as constant, so x^2 vanishes and $2xy$ gives $2x$):

$$\frac{\partial z}{\partial y} = -2y + 2x \implies \frac{\partial^2 z}{\partial y^2} = -2.$$

Adding:

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 2 + (-2) = 0. \quad \blacksquare$$

A function satisfying this is called *harmonic*, and the equation is Laplace's equation. Note that the cross term $2xy$ contributes nothing to either second derivative — it is linear in each variable separately, so it differentiates away.

Answer. $z_{xx} = 2$ and $z_{yy} = -2$, so $z_{xx} + z_{yy} = 0$ as required.

8.3 Higher-order mixed partials

Q.3

Let $f(x, y, z) = x^3y^5z^7 + xy^2 + y^3z - x^2z^2$. Find (a) f_{xz} , (b) f_{yz} , (c) f_{xy} , (d) f_{xyz} , (e) f_{yzx} , (f) f_{xzy} , (g) f_{xxy} , (h) f_{xxz} , (i) f_{yyx} , (j) f_{yzy} .

Solution

Compute the three first partials once, then reuse them throughout.

$$\begin{aligned} f_x &= 3x^2y^5z^7 + y^2 - 2xz^2, \\ f_y &= 5x^3y^4z^7 + 2xy + 3y^2z, \\ f_z &= 7x^3y^5z^6 + y^3 - 2x^2z. \end{aligned}$$

	Differentiate	Result
(a)	f_{xz} f_x w.r.t. z	$21x^2y^5z^6 - 4xz$
(b)	f_{yz} f_y w.r.t. z	$35x^3y^4z^6 + 3y^2$
(c)	f_{xy} f_x w.r.t. y	$15x^2y^4z^7 + 2y$
(d)	f_{xyz} f_{xy} w.r.t. z	$105x^2y^4z^6$
(e)	f_{yzx} f_{yz} w.r.t. x	$105x^2y^4z^6$
(f)	f_{xzy} f_{xz} w.r.t. y	$105x^2y^4z^6$
(g)	f_{xxy} $f_{xx} = 6xy^5z^7 - 2z^2$, then y	$30xy^4z^7$
(h)	f_{xxz} f_{xx} w.r.t. z	$42xy^5z^6 - 4z$
(i)	f_{yyx} $f_{yy} = 20x^3y^3z^7 + 2x + 6yz$, then x	$60x^2y^3z^7 + 2$
(j)	f_{yzy} f_{yz} w.r.t. y	$140x^3y^3z^6 + 6y$

✓ *Check:* Parts (d), (e) and (f) are the same third-order derivative taken in three different orders, and all three give $105x^2y^4z^6$ — exactly as Clairaut's theorem predicts. If any one of them had differed, an arithmetic error would have been located.

Answer. As tabulated. In particular $f_{xyz} = f_{yzx} = f_{xzy} = 105x^2y^4z^6$.

8.4 Interpreting partial derivatives of a cost function

Q.4

Suppose $z = f(x, y)$ represents the cost of constructing a certain structure, where x and y express the labour and material cost respectively. Explain what f_x and f_y represent.

Solution

Both are *marginal costs* — rates of change of total cost — but each with respect to one input while the other is frozen.

f_x : marginal cost with respect to labour. The rate at which total construction cost changes per one-unit increase in labour cost, with material cost held fixed. If $f_x = 1.4$, then spending one more rupee on labour raises the total cost of the structure by about Rs. 1.40 — more than the rupee itself, indicating that labour drags other costs up with it.

f_y : marginal cost with respect to material. The rate at which total cost changes per one-unit increase in material cost, with labour held fixed.

The word “held fixed” carries the meaning. In practice labour and material costs move together, so f_x answers a deliberately artificial but useful question: *if only labour changed, what would happen to the total?* That isolation is what makes the partial derivative the right tool for attributing a cost overrun to one input rather than another.

Answer. f_x is the marginal cost of the structure with respect to labour — the change in total cost per unit change in labour cost, material held constant. f_y is the corresponding marginal cost with respect to material, labour held constant.

8.5 Relative extrema of functions of two variables

Q.6

Find all points where the following functions have relative extrema. (a) $f(x, y) = (y^2 - 4)(e^x - 1)$
 (b) $f(x, y) = x^2 + y^2 + xy - 9x + 1$ (c) $f(x, y) = 4xy - x^4 - y^4$ (d) $f(x, y) = x^2 + y^2 + \frac{32}{xy}$
 (e) $f(x, y) = x e^{y+1}$

Solution

Method throughout: set $f_x = f_y = 0$ to locate critical points, then classify each with $D = f_{xx}f_{yy} - (f_{xy})^2$.

(a) $f_x = (y^2 - 4)e^x$ and $f_y = 2y(e^x - 1)$.

Since $e^x > 0$ always, $f_x = 0$ forces $y^2 = 4$, i.e. $y = \pm 2$. Substituting into $f_y = 0$: $2(\pm 2)(e^x - 1) = 0$ requires $e^x = 1$, so $x = 0$. Critical points: $(0, 2)$ and $(0, -2)$.

$$f_{xx} = (y^2 - 4)e^x, \quad f_{yy} = 2(e^x - 1), \quad f_{xy} = 2ye^x.$$

At both points $f_{xx} = 0$ and $f_{yy} = 0$, while $f_{xy} = \pm 4$:

$$D = 0 \cdot 0 - (\pm 4)^2 = -16 < 0.$$

Answer. (a) Critical points $(0, 2)$ and $(0, -2)$; both are **saddle points** ($D = -16 < 0$). The function has no relative extrema.

(b) $f_x = 2x + y - 9$ and $f_y = 2y + x$.

From $f_y = 0$, $x = -2y$. Substituting into $f_x = 0$:

$$2(-2y) + y - 9 = 0 \implies -3y = 9 \implies y = -3, \quad x = 6.$$

$$f_{xx} = 2, \quad f_{yy} = 2, \quad f_{xy} = 1, \quad D = (2)(2) - 1^2 = 3 > 0, \quad f_{xx} = 2 > 0.$$

$$f(6, -3) = 36 + 9 + 6(-3) - 54 + 1 = -26.$$

Answer. (b) Relative **minimum** at $(6, -3)$, value $f = -26$.

(c) $f_x = 4y - 4x^3$ and $f_y = 4x - 4y^3$.

Setting both to zero gives $y = x^3$ and $x = y^3$. Substituting the first into the second:

$$x = (x^3)^3 = x^9 \implies x^9 - x = 0 \implies x(x^8 - 1) = 0,$$

so $x = 0, 1, -1$ (the real roots), giving critical points $(0, 0)$, $(1, 1)$, $(-1, -1)$.

$$f_{xx} = -12x^2, \quad f_{yy} = -12y^2, \quad f_{xy} = 4, \quad D = 144x^2y^2 - 16.$$

Point	D	f_{xx}	Classification	f
$(0, 0)$	-16	0	saddle point	0
$(1, 1)$	128	-12	relative maximum	2
$(-1, -1)$	128	-12	relative maximum	2

Answer. (c) Relative **maxima** at $(1, 1)$ and $(-1, -1)$, both of value 2; $(0, 0)$ is a saddle point.

(d) Write $f = x^2 + y^2 + 32x^{-1}y^{-1}$, so

$$f_x = 2x - \frac{32}{x^2y}, \quad f_y = 2y - \frac{32}{xy^2}.$$

Setting each to zero:

$$2x^3y = 32 \Rightarrow x^3y = 16, \quad 2xy^3 = 32 \Rightarrow xy^3 = 16.$$

Dividing one by the other gives $x^2/y^2 = 1$, so $y = \pm x$.

If $y = x$: $x^4 = 16 \Rightarrow x = \pm 2$, giving $(2, 2)$ and $(-2, -2)$. If $y = -x$: $-x^4 = 16$, which has no real solution.

$$f_{xx} = 2 + \frac{64}{x^3y}, \quad f_{yy} = 2 + \frac{64}{xy^3}, \quad f_{xy} = \frac{32}{x^2y^2}.$$

At $(2, 2)$: $f_{xx} = 2 + \frac{64}{16} = 6$, $f_{yy} = 6$, $f_{xy} = \frac{32}{16} = 2$,

$$D = 36 - 4 = 32 > 0, \quad f_{xx} = 6 > 0 \Rightarrow \text{minimum},$$

$$f(2, 2) = 4 + 4 + \frac{32}{4} = 16.$$

At $(-2, -2)$ the same values arise ($x^3y = 16$ and $x^2y^2 = 16$ again), so $D = 32 > 0$, $f_{xx} = 6 > 0$, and $f(-2, -2) = 16$.

Answer. (d) Relative **minima** at $(2, 2)$ and $(-2, -2)$, both of value 16.

(e) $f_x = e^{y+1}$ and $f_y = xe^{y+1}$.

Since $e^{y+1} > 0$ for every y , f_x is *never* zero. There are no critical points, hence nothing to classify.

Answer. (e) No critical points ($f_x = e^{y+1} \neq 0$ everywhere), so the function has **no relative extrema**.

Where marks are lost

$D < 0$ means saddle point — a genuine, complete answer, not a failure. And $D = 0$ means the test is *inconclusive*, not that the point is a saddle; in that case the test simply cannot decide and other reasoning is needed.

8.6 A production function — and a second defective question

Q.7

Suppose a production function P is given by

$$P = f(l, k) = 0.8l^3 - 0.02l^2 + 2.5k^2 - 0.05k^3.$$

Find the values of l and k so as to maximise output P .

Solution

As printed, no maximum exists. Take the partials:

$$\frac{\partial P}{\partial l} = 2.4l^2 - 0.04l = l(2.4l - 0.04), \quad \frac{\partial P}{\partial k} = 5k - 0.15k^2 = k(5 - 0.15k).$$

Setting them to zero gives $l = 0$ or $l = \frac{0.04}{2.4} = \frac{1}{60}$, and $k = 0$ or $k = \frac{5}{0.15} = \frac{100}{3}$.

Now the second-order condition in l :

$$\frac{\partial^2 P}{\partial l^2} = 4.8l - 0.04, \quad \text{at } l = \frac{1}{60}: \quad 4.8\left(\frac{1}{60}\right) - 0.04 = 0.08 - 0.04 = 0.04 > 0.$$

The critical point in l is a *minimum*, not a maximum. And more decisively, the leading term $0.8l^3$ is positive and cubic, so

$$P \rightarrow +\infty \quad \text{as } l \rightarrow \infty.$$

An unbounded function has no maximum. The question as printed cannot be answered.

Answer. As printed, P is unbounded above in l (the $+0.8l^3$ term dominates), so **no maximising (l, k) exists**. The stationary point $l = \frac{1}{60}$ is a minimum in l , since $P_{ll} = 0.04 > 0$ there.

The intended problem. A production function that can be maximised needs a *negative* cubic term, and the standard form of this exercise is

$$P = f(l, k) = 0.8l^2 - 0.02l^3 + 2.5k^2 - 0.05k^3,$$

where the exponents on the first two terms are interchanged. Then

$$\frac{\partial P}{\partial l} = 1.6l - 0.06l^2 = l(1.6 - 0.06l) = 0 \implies l = 0 \quad \text{or} \quad l = \frac{1.6}{0.06} = \frac{80}{3} \approx 26.67,$$

$$\frac{\partial P}{\partial k} = 5k - 0.15k^2 = k(5 - 0.15k) = 0 \implies k = 0 \quad \text{or} \quad k = \frac{5}{0.15} = \frac{100}{3} \approx 33.33.$$

Classifying the non-trivial point:

$$\frac{\partial^2 P}{\partial l^2} = 1.6 - 0.12l \Big|_{l=80/3} = 1.6 - 3.2 = -1.6 < 0,$$

$$\frac{\partial^2 P}{\partial k^2} = 5 - 0.3k \Big|_{k=100/3} = 5 - 10 = -5 < 0, \quad P_{lk} = 0,$$

$$D = (-1.6)(-5) - 0^2 = 8 > 0, \quad P_{ll} < 0 \implies \text{relative maximum.}$$

Maximum output:

$$P = 0.8\left(\frac{80}{3}\right)^2 - 0.02\left(\frac{80}{3}\right)^3 + 2.5\left(\frac{100}{3}\right)^2 - 0.05\left(\frac{100}{3}\right)^3 \approx 189.63 + 925.93 = 1115.56.$$

Answer. For the intended function $P = 0.8l^2 - 0.02l^3 + 2.5k^2 - 0.05k^3$: output is maximised at $l = \frac{80}{3} \approx 26.67$ and $k = \frac{100}{3} \approx 33.33$, with $D = 8 > 0$ and $P_{ll} = -1.6 < 0$ confirming a maximum; $P_{\max} \approx 1115.6$.

Where marks are lost

Never stop at “ $P_l = 0$ and $P_k = 0$, therefore maximum”. The second-order test is what distinguishes a maximum from a minimum or a saddle, and in this question it is what reveals that the printed function has no maximum at all. Examiners award the second-order working separately.

8.7 Maximising profit in two variables**Q.8**

The profit of a certain company is given by

$$P(x, y) = 1200 + 80x - 2x^2 + 100y - y^2,$$

where x and y represent the cost of a unit of labour and a unit of goods respectively. Find x and y that maximise the profit, and find the maximum profit.

Solution

The variables separate — there is no xy term — so each can be optimised independently. Setting the first partials to zero:

$$\begin{aligned}\frac{\partial P}{\partial x} &= 80 - 4x = 0 \implies x = 20, \\ \frac{\partial P}{\partial y} &= 100 - 2y = 0 \implies y = 50.\end{aligned}$$

Second-order test:

$$\begin{aligned}P_{xx} &= -4, & P_{yy} &= -2, & P_{xy} &= 0, \\ D &= (-4)(-2) - 0^2 = 8 > 0 \quad \text{and} \quad P_{xx} = -4 < 0 \implies \text{relative maximum.}\end{aligned}$$

Maximum profit:

$$\begin{aligned}P(20, 50) &= 1200 + 80(20) - 2(400) + 100(50) - (50)^2 \\ &= 1200 + 1600 - 800 + 5000 - 2500 = 4500.\end{aligned}$$

✓ *Check:* Perturbing either variable lowers P : $P(21, 50) = 4498$ and $P(20, 51) = 4499$. Consistent with a maximum at $(20, 50)$.

Answer. Profit is maximised at $x = 20$ and $y = 50$, giving a maximum profit of **4,500**.

8.8 Maximising revenue with an interaction term**Q.9**

The revenue in thousands of dollars from selling x spas and y solar heaters is

$$R(x, y) = 15 + 80x + 81y - 2x^2 - 3y^2 - 4xy.$$

Find the quantity of each that must be sold to generate maximum revenue, and find the maximum revenue.

Solution

Here the $-4xy$ term couples the two products, so the equations must be solved *simultaneously*.

$$\frac{\partial R}{\partial x} = 80 - 4x - 4y = 0 \implies x + y = 20, \quad (1)$$

$$\frac{\partial R}{\partial y} = 81 - 6y - 4x = 0 \implies 4x + 6y = 81. \quad (2)$$

From (1), $x = 20 - y$. Substituting into (2):

$$4(20 - y) + 6y = 81 \implies 80 + 2y = 81 \implies y = 0.5,$$

$$x = 20 - 0.5 = 19.5.$$

Second-order test:

$$R_{xx} = -4, \quad R_{yy} = -6, \quad R_{xy} = -4,$$

$$D = (-4)(-6) - (-4)^2 = 24 - 16 = 8 > 0, \quad R_{xx} = -4 < 0 \implies \text{maximum}.$$

Maximum revenue:

$$\begin{aligned} R(19.5, 0.5) &= 15 + 80(19.5) + 81(0.5) - 2(19.5)^2 - 3(0.5)^2 - 4(19.5)(0.5) \\ &= 15 + 1560 + 40.5 - 760.5 - 0.75 - 39 \\ &= 815.25. \end{aligned}$$

Revenue is measured in thousands of dollars, so this is \$815,250.

A practical note. Spas and heaters are sold in whole units, so the exact optimum (19.5, 0.5) is not attainable. Testing the four nearest integer points:

$$R(20, 0) = 815.00, \quad R(19, 1) = 815.00, \quad R(20, 1) = 813.00, \quad R(19, 0) = 813.00.$$

The best integer plans are 20 spas with no heaters, or 19 spas with 1 heater, each giving \$815,000 — \$250 below the continuous optimum.

Answer. Revenue is maximised at $x = 19.5$ spas and $y = 0.5$ heaters, with $R_{\max} = 815.25$ thousand dollars (\$815,250). Restricted to whole units, (20, 0) or (19, 1) both give \$815,000.

Where marks are lost

With a cross term present, $D = f_{xx}f_{yy} - (f_{xy})^2$ can go negative even when both f_{xx} and f_{yy} are negative. Here $24 - 16 = 8$ only just clears zero. Never skip the $(f_{xy})^2$ subtraction on the grounds that both second derivatives “look right”.

Optimization

Source: *Business Mathematics* 5405/1429, §9.9 SELF ASSESSMENT QUESTIONS, pp. 205–206.

What the two derivatives tell you

	f' — first derivative	f'' — second derivative
Zero at	stationary points	possible inflection points
Positive means	f increasing	f concave up (holds water)
Negative means	f decreasing	f concave down (spills water)

First derivative test: f' changing $+$ $\rightarrow$ $-$ gives a maximum, $- \rightarrow +$ a minimum, no change gives neither.

Second derivative test: at a stationary point, $f'' < 0 \Rightarrow$ maximum, $f'' > 0 \Rightarrow$ minimum, $f'' = 0 \Rightarrow$ inconclusive.

An inflection point needs f'' to change sign, not merely to vanish.

The four business functions

$$R(x) = x \cdot p(x), \quad P(x) = R(x) - C(x), \quad \bar{C}(x) = \frac{C(x)}{x},$$

and their marginals are the derivatives $MR = R'$, $MC = C'$, $MP = P'$. Profit is maximised where $P' = 0$, i.e. where $MR = MC$, provided $P'' < 0$.

9.1 Increase, decrease, concavity and inflection

Q.(a)

Find the intervals on which f is increasing, decreasing, concave up and concave down, and locate any points of inflection. (i) $f(x) = x^2 - 3x + 8$ (ii) $f(x) = \frac{x^4}{3} - \frac{x}{3}$ (iii) $f(x) = \frac{x}{x^2 + 2}$
(iv) $f(x) = x^4 - 5x^3 + 9x^2$

Solution

(i) $f'(x) = 2x - 3$, zero at $x = \frac{3}{2}$. Since $f' < 0$ to the left and $f' > 0$ to the right:

$$\text{decreasing on } \left(-\infty, \frac{3}{2}\right), \quad \text{increasing on } \left(\frac{3}{2}, \infty\right).$$

$f''(x) = 2 > 0$ everywhere, so the curve is concave up throughout and has *no* inflection point. The stationary point $x = \frac{3}{2}$ is a minimum, $f(\frac{3}{2}) = \frac{9}{4} - \frac{9}{2} + 8 = \frac{23}{4} = 5.75$.

(ii) $f(x) = \frac{1}{3}(x^4 - x)$, so

$$f'(x) = \frac{4x^3 - 1}{3} = 0 \implies x^3 = \frac{1}{4} \implies x = 4^{-1/3} \approx 0.630.$$

decreasing on $(-\infty, 0.630)$, increasing on $(0.630, \infty)$.

$$f''(x) = \frac{12x^2}{3} = 4x^2 \geq 0 \quad \text{for all } x.$$

f'' vanishes at $x = 0$ but does *not* change sign there, so the curve is concave up everywhere and has no inflection point.

(iii) Quotient rule:

$$f'(x) = \frac{(x^2 + 2) - x(2x)}{(x^2 + 2)^2} = \frac{2 - x^2}{(x^2 + 2)^2}.$$

The denominator is always positive, so the sign of f' is the sign of $2 - x^2$: zero at $x = \pm\sqrt{2}$.

increasing on $(-\sqrt{2}, \sqrt{2})$, decreasing on $(-\infty, -\sqrt{2})$ and $(\sqrt{2}, \infty)$.

Differentiating again and simplifying:

$$f''(x) = \frac{2x(x^2 - 6)}{(x^2 + 2)^3}.$$

Zeros at $x = 0$ and $x = \pm\sqrt{6}$; the denominator is positive, so the sign is that of $2x(x^2 - 6)$:

Interval	$2x$	$x^2 - 6$	f''
$x < -\sqrt{6}$	−	+	− concave down
$-\sqrt{6} < x < 0$	−	−	+ concave up
$0 < x < \sqrt{6}$	+	−	− concave down
$x > \sqrt{6}$	+	+	+ concave up

f'' changes sign at all three zeros, so there are inflection points at $x = -\sqrt{6}$, $x = 0$ and $x = \sqrt{6}$, with $f(0) = 0$ and $f(\pm\sqrt{6}) = \pm\frac{\sqrt{6}}{8}$.

(iv) $f'(x) = 4x^3 - 15x^2 + 18x = x(4x^2 - 15x + 18)$.

The quadratic factor has discriminant $225 - 288 = -63 < 0$ and positive leading coefficient, so it is positive for every x and never contributes a root. Hence $f' = 0$ only at $x = 0$, and the sign of f' is the sign of x :

decreasing on $(-\infty, 0)$, increasing on $(0, \infty)$,

with a relative minimum at $x = 0$, $f(0) = 0$.

$$f''(x) = 12x^2 - 30x + 18 = 6(2x^2 - 5x + 3) = 6(2x - 3)(x - 1),$$

zero at $x = 1$ and $x = \frac{3}{2}$:

concave up on $(-\infty, 1)$ and $(\frac{3}{2}, \infty)$; concave down on $(1, \frac{3}{2})$.

Both are genuine inflection points, at $(1, 5)$ and $(\frac{3}{2}, \frac{135}{16}) = (1.5, 8.4375)$.

Answer. (i) decreasing then increasing about $x = \frac{3}{2}$; concave up everywhere; no inflection.
 (ii) decreasing then increasing about $x = 4^{-1/3} \approx 0.630$; concave up everywhere; no inflection.
 (iii) increasing on $(-\sqrt{2}, \sqrt{2})$, decreasing outside; inflections at $x = -\sqrt{6}$, 0 , $\sqrt{6}$. (iv) decreasing on $(-\infty, 0)$, increasing on $(0, \infty)$; concave down only on $(1, \frac{3}{2})$; inflections at $(1, 5)$ and $(1.5, 8.4375)$.

Where marks are lost

In (ii), $f''(0) = 0$ but $f'' = 4x^2$ is non-negative on both sides, so there is *no* inflection at $x = 0$. “ $f'' = 0$ ” identifies a candidate; only a sign change confirms it.

9.2 Relative extrema by both derivative tests**Q.(b)**

Use both the first and the second derivative test to find the relative extrema of f . (i) $f(x) = x^4 - 2x^3 + 4x^2$ (ii) $f(x) = x^2(x - 2)^2$ (iii) $f(x) = 3x + 2x^{1/3}$ (iv) $f(x) = \frac{x^2}{x^4 + x^2 - 1}$
(v) $f(x) = \frac{x + 4}{x - 3}$

Solution

(i) $f'(x) = 4x^3 - 6x^2 + 8x = 2x(2x^2 - 3x + 4)$.

The quadratic has discriminant $9 - 32 = -23 < 0$, so it is always positive and contributes no roots. Thus $x = 0$ is the only critical point.

First derivative test: the sign of f' is the sign of $2x$, so f' changes $- \rightarrow +$ at $x = 0$: a relative minimum.

Second derivative test: $f''(x) = 12x^2 - 12x + 8$, and $f''(0) = 8 > 0$: minimum. Both agree. $f(0) = 0$.

(ii) Recognise $f = [x(x - 2)]^2 = (x^2 - 2x)^2$, then use the chain rule:

$$f'(x) = 2(x^2 - 2x)(2x - 2) = 4x(x - 2)(x - 1).$$

Critical points $x = 0, 1, 2$.

First derivative test:

Interval	$4x$	$(x - 2)$	$(x - 1)$	f'	f
$x < 0$	−	−	−	−	decreasing
$0 < x < 1$	+	−	−	+	increasing
$1 < x < 2$	+	−	+	−	decreasing
$x > 2$	+	+	+	+	increasing

So $x = 0$ is a minimum, $x = 1$ a maximum and $x = 2$ a minimum.

Second derivative test: $f''(x) = 4(3x^2 - 6x + 2)$, so

$$f''(0) = 8 > 0 \text{ (min)}, \quad f''(1) = -4 < 0 \text{ (max)}, \quad f''(2) = 8 > 0 \text{ (min)}.$$

Values: $f(0) = 0$, $f(1) = 1$, $f(2) = 0$. The two tests agree throughout.

(iii) $f(x) = 3x + 2x^{1/3}$, so

$$f'(x) = 3 + \frac{2}{3}x^{-2/3} = 3 + \frac{2}{3x^{2/3}}.$$

Since $x^{2/3} > 0$ for every $x \neq 0$, $f' > 3 > 0$ everywhere it is defined, and f' is undefined (infinite) at $x = 0$. There is no point where $f' = 0$, and f' does not change sign across $x = 0$.

Therefore f is strictly increasing on its whole domain and has **no relative extrema**. The second derivative test is not applicable: it requires a stationary point, and there is none.

(iv) $f(x) = \frac{x^2}{x^4 + x^2 - 1}$. By the quotient rule,

$$\begin{aligned} f'(x) &= \frac{2x(x^4 + x^2 - 1) - x^2(4x^3 + 2x)}{(x^4 + x^2 - 1)^2} = \frac{2x^5 + 2x^3 - 2x - 4x^5 - 2x^3}{(x^4 + x^2 - 1)^2} \\ &= \frac{-2x^5 - 2x}{(x^4 + x^2 - 1)^2} = \frac{-2x(x^4 + 1)}{(x^4 + x^2 - 1)^2}. \end{aligned}$$

The denominator is positive and $x^4 + 1 > 0$ always, so the sign of f' is the sign of $-2x$: positive for $x < 0$, negative for $x > 0$.

First derivative test: f' changes $+$ $\rightarrow$ $-$ at $x = 0$, so there is a relative **maximum** at $x = 0$, with

$$f(0) = \frac{0}{0 + 0 - 1} = 0.$$

The domain excludes the points where $x^4 + x^2 - 1 = 0$: putting $u = x^2$, $u^2 + u - 1 = 0$ gives $u = \frac{-1 \pm \sqrt{5}}{2} \approx 0.618$, so $x \approx \pm 0.786$. Between those two vertical asymptotes the denominator is negative, so $f \leq 0$ there — consistent with $f(0) = 0$ being the largest value on that interval.

(v) $f(x) = \frac{x+4}{x-3}$:

$$f'(x) = \frac{(x-3) - (x+4)}{(x-3)^2} = \frac{-7}{(x-3)^2}.$$

This is negative for every $x \neq 3$ and never zero. There are no critical points, so f has **no relative extrema**; it is strictly decreasing on $(-\infty, 3)$ and on $(3, \infty)$ separately.

$$f''(x) = \frac{14}{(x-3)^3} : \quad \text{concave down for } x < 3, \quad \text{concave up for } x > 3.$$

There is a vertical asymptote at $x = 3$ and a horizontal asymptote at $y = 1$.

Answer. (i) Relative minimum at $(0, 0)$. (ii) Minima at $(0, 0)$ and $(2, 0)$, maximum at $(1, 1)$. (iii) None — f is strictly increasing. (iv) Relative maximum at $(0, 0)$. (v) None — f is strictly decreasing on each branch.

Where marks are lost

In (iii) and (v), “no relative extrema” is the full answer. A function with no stationary point has nothing to classify, and the second derivative test cannot be applied. Do not invent an extremum at a point where the derivative is merely undefined ((iii), $x = 0$) or where the function itself is undefined ((v), $x = 3$).

9.3 Curve sketching

Q.(c)

Sketch the graph of the functions (i) $f(x) = 2x^3 - 3x^2 - 12x + 1$ (ii) $f(x) = \frac{3x^2}{x^2 + 5}$ (iii)

$f(x) = x - \frac{1}{x}$ (iv) $f(x) = \frac{x-1}{x+1}$ (v) $f(x) = x^3$

Solution

A sketch is a report on four things: stationary points, concavity, asymptotes, and intercepts. Take (i) and (ii) in full.

(i) $f(x) = 2x^3 - 3x^2 - 12x + 1$.

$$f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2) = 6(x - 2)(x + 1),$$

so stationary points at $x = -1$ and $x = 2$:

$$f(-1) = -2 - 3 + 12 + 1 = 8, \quad f(2) = 16 - 12 - 24 + 1 = -19.$$

$$f''(x) = 12x - 6, \quad f''(-1) = -18 < 0 \text{ (max)}, \quad f''(2) = 18 > 0 \text{ (min)}.$$

$f'' = 0$ at $x = \frac{1}{2}$, where it changes sign: an inflection point at $(\frac{1}{2}, -5.5)$. So: increasing, then decreasing between -1 and 2 , then increasing again; concave down for $x < \frac{1}{2}$, concave up after.

(ii) $f(x) = \frac{3x^2}{x^2 + 5}$.

$$f'(x) = \frac{6x(x^2 + 5) - 3x^2(2x)}{(x^2 + 5)^2} = \frac{30x}{(x^2 + 5)^2},$$

zero only at $x = 0$, changing $- \rightarrow +$: a minimum at $(0, 0)$.

$$f''(x) = \frac{150 - 90x^2}{(x^2 + 5)^3},$$

zero at $x = \pm\sqrt{5/3} \approx \pm 1.291$, giving inflection points at $(\pm 1.291, 0.75)$. As $x \rightarrow \pm\infty$, $f \rightarrow 3$: a horizontal asymptote at $y = 3$, approached from below. There is no vertical asymptote, since $x^2 + 5$ never vanishes.

(iii) $f(x) = x - \frac{1}{x}$ has $f' = 1 + \frac{1}{x^2} > 0$ always, so it is increasing on each branch, with a vertical asymptote at $x = 0$ and the slant asymptote $y = x$.

(iv) $f(x) = \frac{x-1}{x+1}$ has $f' = \frac{2}{(x+1)^2} > 0$, so it is increasing on each branch, with a vertical asymptote at $x = -1$, a horizontal asymptote at $y = 1$, and intercepts $(1, 0)$ and $(0, -1)$.

(v) $f(x) = x^3$ has $f' = 3x^2 \geq 0$ (increasing everywhere, with a horizontal tangent but no extremum at $x = 0$) and $f'' = 6x$, giving an inflection point at the origin.

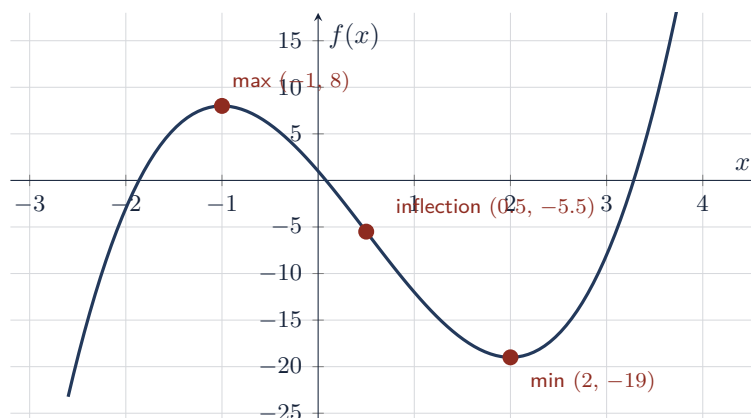


Figure 9.1: Q.(c)(i): $f(x) = 2x^3 - 3x^2 - 12x + 1$, showing the maximum, the minimum, and the inflection point where concavity reverses.

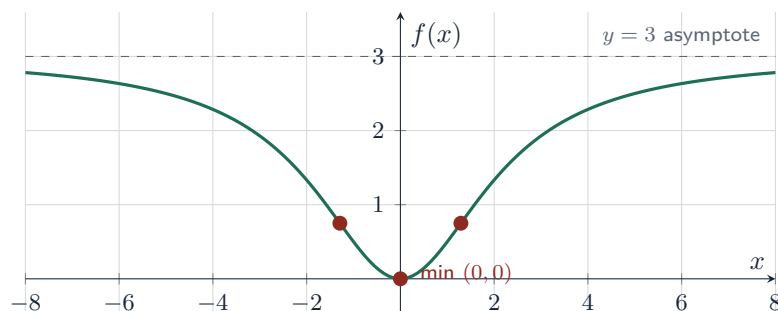


Figure 9.2: Q.(c)(ii): $f(x) = 3x^2/(x^2 + 5)$, with its minimum at the origin, inflection points at $x = \pm\sqrt{5/3}$, and the horizontal asymptote $y = 3$.

Answer. (i) Maximum $(-1, 8)$, minimum $(2, -19)$, inflection $(0.5, -5.5)$. (ii) Minimum $(0, 0)$, inflections at $(\pm 1.291, 0.75)$, horizontal asymptote $y = 3$. (iii) Increasing on each branch; asymptotes $x = 0$ and $y = x$. (iv) Increasing on each branch; asymptotes $x = -1$ and $y = 1$. (v) Increasing everywhere; inflection at the origin.

9.4 Where profit is increasing

Q.(d)

A manufacturer sells a product with cost and revenue functions

$$C(x) = 2.2x - 0.0002x^2, \quad R(x) = 4.5x - 0.004x^2, \quad 0 \leq x \leq 1200,$$

where x is the number of items sold. Find the interval on which the profit function is increasing.

Solution

Build the profit function first, and collect like terms carefully — the subtraction of a negative x^2 term is where errors enter:

$$\begin{aligned} P(x) &= R(x) - C(x) \\ &= (4.5x - 0.004x^2) - (2.2x - 0.0002x^2) \\ &= 4.5x - 0.004x^2 - 2.2x + 0.0002x^2 \\ &= 2.3x - 0.0038x^2. \end{aligned}$$

Profit is increasing where $P'(x) > 0$:

$$P'(x) = 2.3 - 0.0076x > 0 \iff x < \frac{2.3}{0.0076} = 302.63.$$

Since the domain is $0 \leq x \leq 1200$:

$$P \text{ increasing on } [0, 302.63), \quad P \text{ decreasing on } (302.63, 1200].$$

$P'' = -0.0076 < 0$, so the turning point is a maximum:

$$P(302.63) = 2.3(302.63) - 0.0038(302.63)^2 = 696.05 - 348.03 = 348.03.$$

In whole units, output of about 303 items maximises profit at roughly \$348.

Answer. Profit is increasing on $0 \leq x < 302.63$, i.e. up to about **303 units**, after which it declines. Maximum profit $\approx \$348.03$.

Where marks are lost

$-(-0.0002x^2) = +0.0002x^2$. Dropping that sign gives $P = 2.3x - 0.0042x^2$ and a turning point at 274 instead of 303. Write out the subtraction line in full rather than doing it in your head.

9.5 When is marginal revenue increasing?

Q.(e)

The demand function is $p(x) = 300 - 2x$. Determine when marginal revenue is increasing.

Solution

Revenue is price times quantity:

$$R(x) = x p(x) = x(300 - 2x) = 300x - 2x^2.$$

Marginal revenue is the first derivative:

$$MR(x) = R'(x) = 300 - 4x.$$

Marginal revenue is increasing where *its* derivative is positive, i.e. where $R'' > 0$:

$$\frac{d}{dx}(MR) = R''(x) = -4.$$

This is a negative constant — it is never positive. Therefore marginal revenue is **never increasing**: it is a decreasing linear function of x , falling by \$4 for each extra unit sold, on the whole domain.

What is true instead. $MR > 0$ (revenue itself still rising) for $x < 75$; $MR = 0$ at $x = 75$, where revenue peaks at $R(75) = 300(75) - 2(75)^2 = 22500 - 11250 = 11,250$; and $MR < 0$ beyond that, where selling more actually reduces revenue.

Answer. Marginal revenue $MR = 300 - 4x$ has $R'' = -4 < 0$ everywhere, so marginal revenue is **never increasing** — it decreases at a constant rate of 4 per unit. Revenue itself increases only while $MR > 0$, that is for $x < 75$.

Where marks are lost

“Revenue increasing” and “marginal revenue increasing” are different questions. The first asks about $R' > 0$; the second about $R'' > 0$. This question asks the second, and the answer — “never” — is a legitimate one.

9.6 Minimising average cost

Q.(f)

The total cost c of producing x units of a product is $c = 0.03x^2 + 8x + 300$. At what level of output will the average cost per unit be minimum?

Solution

Average cost is total cost divided by output:

$$\bar{c}(x) = \frac{0.03x^2 + 8x + 300}{x} = 0.03x + 8 + \frac{300}{x} = 0.03x + 8 + 300x^{-1}.$$

Differentiate and set to zero:

$$\bar{c}'(x) = 0.03 - \frac{300}{x^2} = 0 \implies x^2 = \frac{300}{0.03} = 10,000 \implies x = 100$$

(taking the positive root, since output cannot be negative).

Second derivative test:

$$\bar{c}''(x) = \frac{600}{x^3}, \quad \bar{c}''(100) = \frac{600}{10^6} > 0 \implies \text{minimum.}$$

$$\bar{c}(100) = 0.03(100) + 8 + \frac{300}{100} = 3 + 8 + 3 = 14.$$

✓ *Check:* $\bar{c}(90) = 2.7 + 8 + 3.333 = 14.03$ and $\bar{c}(110) = 3.3 + 8 + 2.727 = 14.03$. Both exceed 14, confirming the minimum.

Worth noting. At the minimum, marginal cost equals average cost: $c'(x) = 0.06x + 8$, so $c'(100) = 14 = \bar{c}(100)$. That identity always holds at the minimum of average cost, and makes a good check.

Answer. Average cost is minimised at an output of **100 units**, where the average cost is **\$14** per unit.

9.7 Maximising profit from demand and cost

Q.(g)

The demand p and cost function c for a product are

$$p = 52 - 0.02x, \quad c(x) = 400 + 40x.$$

At what level of output will profit be maximised? At what price does this profit arise, and what is the profit?

Solution

Revenue is price times quantity:

$$R(x) = x p = x(52 - 0.02x) = 52x - 0.02x^2.$$

Profit:

$$P(x) = R(x) - c(x) = 52x - 0.02x^2 - 400 - 40x = 12x - 0.02x^2 - 400.$$

$$P'(x) = 12 - 0.04x = 0 \implies x = \frac{12}{0.04} = 300.$$

$$P''(x) = -0.04 < 0 \implies \text{maximum.}$$

Price at that output:

$$p = 52 - 0.02(300) = 52 - 6 = \$46.$$

Profit:

$$P(300) = 12(300) - 0.02(300)^2 - 400 = 3600 - 1800 - 400 = \$1400.$$

✓ *Check:* Via $MR = MC$: $MR = 52 - 0.04x$ and $MC = 40$, so $52 - 0.04x = 40$ gives $x = 300$ — the same output, reached by the economic rule rather than the calculus one. And $R(300) = 15600 - 1800 = 13,800$, $c(300) = 400 + 12000 = 12,400$, difference = 1,400.

Answer. Profit is maximised at an output of **300 units**, at a price of **\$46** per unit, giving a maximum profit of **\$1,400**.

Where marks are lost

Maximum profit is not maximum revenue. Revenue here peaks at $x = 1300$ (where $52 - 0.04x = 0$), far beyond the profit-maximising 300. Optimise the function the question names.

9.8 Pricing a cable subscription

Q.(h)

A TV cable company has 500 subscribers paying \$20 per month. There will be 200 more subscribers for each \$0.60 decrease in the monthly fee. At what rate will maximum revenue be obtained, and what will this revenue be?

Solution

Let n be the number of \$0.60 reductions applied. Then

Monthly fee	$20 - 0.6n$ dollars
Number of subscribers	$500 + 200n$

Revenue is fee times subscribers:

$$\begin{aligned} R(n) &= (20 - 0.6n)(500 + 200n) \\ &= 10000 + 4000n - 300n - 120n^2 \\ &= 10000 + 3700n - 120n^2. \end{aligned}$$

$$R'(n) = 3700 - 240n = 0 \implies n = \frac{3700}{240} = 15.4167.$$

$$R''(n) = -240 < 0 \implies \text{maximum.}$$

At that n :

$$\text{fee} = 20 - 0.6(15.4167) = 20 - 9.25 = \$10.75,$$

$$\text{subscribers} = 500 + 200(15.4167) = 3583.3,$$

$$R = 10000 + 3700(15.4167) - 120(15.4167)^2 = 10000 + 57,041.67 - 28,520.83 = \$38,520.83.$$

The whole-number answer. n must be a whole number of \$0.60 reductions, and subscribers come in whole people. Testing the two neighbours:

n	Fee	Subscribers	Revenue
15	\$11.00	3,500	\$38,500
16	\$10.40	3,700	\$38,480

So the best attainable plan is $n = 15$: a fee of \$11.00 per month.

Answer. The continuous optimum is a monthly rate of **\$10.75** with $\approx 3,583$ subscribers, giving revenue of **\$38,520.83**. Restricted to whole \$0.60 reductions, the best rate is **\$11.00** per month with 3,500 subscribers, giving **\$38,500** per month.

Where marks are lost

Define your variable explicitly — “ n = number of \$0.60 decreases” — and write both the fee and the subscriber count in terms of it before multiplying. Setting up R in terms of the fee directly is possible but invites sign errors, and most lost marks in this question are lost before any calculus begins.

Formula Sheet

Everything the nine units require, on four pages. Learn the *shape* of each rule and where it applies; the arithmetic is the easy part.

A.1 Probability (Units 1–2)

Rule	Formula	Use it when
Complement	$P(A') = 1 - P(A)$	“at least one”, “not”
Addition (general)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	“or”, events overlap
Addition (exclusive)	$P(A \cup B) = P(A) + P(B)$	“or”, $A \cap B = \emptyset$
Multiplication (indep.)	$P(A \cap B) = P(A)P(B)$	“and”, with replacement
Multiplication (dep.)	$P(A \cap B) = P(A)P(B A)$	“and”, without replacement
Conditional	$P(A B) = \frac{P(A \cap B)}{P(B)}$	“given that”
Independence test	$P(A \cap B) = P(A)P(B)$	to prove or disprove it

Distribution	Formula	Recognise it by
Binomial	$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$, $E(X) = np$	n fixed trials, constant p , <i>with</i> replacement
Hypergeometric	$P(X = x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$	sampling from a finite bag <i>without</i> replacement
Any discrete	$\sum_x P(X = x) = 1$, $E(X) = \sum_x x P(X = x)$	always — use as a check

A.2 Equations, inequalities, straight lines (Units 3–4)

Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$; discriminant $b^2 - 4ac > 0$ two roots, $= 0$ one, < 0 none real
Completing the square	$x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \frac{b^2}{4}$
Inequalities	multiplying or dividing by a <i>negative</i> reverses the sign
Absolute value (equation)	$ u = v \iff u = v \text{ or } u = -v$
Absolute value ($\leq$)	$ u \leq a \iff -a \leq u \leq a$ (one interval)
Absolute value ($\geq$)	$ u \geq a \iff u \geq a \text{ or } u \leq -a$ (two intervals)
Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint	$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$
Slope	$m = \frac{y_2 - y_1}{x_2 - x_1}$
Point–slope / slope–intercept	$y - y_1 = m(x - x_1)$; $y = mx + c$
Parallel / perpendicular	$m_1 = m_2$; $m_1 m_2 = -1$ (not for vertical lines)
Linear cost	$C(x) = (\text{variable per unit})x + (\text{fixed cost})$
Equilibrium	set demand = supply and solve for q , then substitute for p

A.3 Matrices and determinants (Units 5–6)

Conformability	$A_{m \times n} B_{n \times p}$ is defined and is $m \times p$
Transpose rules	$(A^t)^t = A$, $(A + B)^t = A^t + B^t$, $(AB)^t = B^t A^t$
Inverse rules	$(A^{-1})^{-1} = A$, $(AB)^{-1} = B^{-1} A^{-1}$
2×2 determinant	$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$
2×2 inverse	$\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$
Cofactor	$C_{ij} = (-1)^{i+j} M_{ij}$; sign pattern $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$
Determinant	$\det A = \sum_j a_{ij} C_{ij}$ along any row or column
Adjoint	$\text{adj } A = [C_{ij}]^t$ — <i>transpose</i> the cofactor matrix
Inverse	$A^{-1} = \frac{1}{\det A} \text{adj } A$, requires $\det A \neq 0$
Cramer's rule	$x_k = \frac{D_k}{D}$, $D = \det A$, D_k replaces column k by b
$\det = 0$ means	no unique solution: either none, or infinitely many
Product rule for det	$\det(AB) = \det A \cdot \det B$

Determinant shortcuts

Expand along the row or column with most zeros. A common factor in a row or column comes outside. Two identical or proportional rows (or columns) force $\det = 0$. Adding a multiple of one row to another leaves $\det$ unchanged.

A.4 Calculus (Units 7–9)

Rule	Formula	Note
Power	$\frac{d}{dx} x^n = nx^{n-1}$	rewrite roots and reciprocals as powers first
Constant multiple	$\frac{d}{dx} [cf] = cf'$	
Sum	$(u + v)' = u' + v'$	
Product	$(uv)' = u'v + uv'$	
Quotient	$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$	derivative of the <i>top</i> first
Chain	$\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$	outer first, then $\times$ inner
Exponential	$\frac{d}{dx} e^u = e^u u'$	
Logarithm	$\frac{d}{dx} \ln u = \frac{u'}{u}$	

Limits at infinity	divide by the highest power of x in the denominator
$\frac{0}{0}$ form	factorise and cancel, then substitute again
Limit exists at a	$\lim_{x \rightarrow a^-} f = \lim_{x \rightarrow a^+} f$
Partial derivative	differentiate in one variable, treat the other as a constant
Clairaut	$f_{xy} = f_{yx}$ — use as a free check on mixed partials
Stationary point	$f'(x) = 0$ (or $f_x = f_y = 0$ in two variables)
Second-derivative test (1 var)	$f'' < 0$ max, $f'' > 0$ min, $f'' = 0$ inconclusive
Second-derivative test (2 var)	$D = f_{xx}f_{yy} - (f_{xy})^2$; $D > 0$ with $f_{xx} < 0$ max, with $f_{xx} > 0$ min; $D < 0$ saddle; $D = 0$ inconclusive
Inflection point	$f'' = 0$ and f'' changes sign
Revenue	$R(x) = x \cdot p(x)$
Profit	$P(x) = R(x) - C(x)$; maximised where $MR = MC$ and $P'' < 0$
Average cost	$\bar{C}(x) = \frac{C(x)}{x}$; at its minimum, $MC = \bar{C}$

Method in the Exam Hall

B.1 The eight recurring mistakes

Collected from every *Where marks are lost* box in this document, in the order the units come:

1. **Using $P(A) + P(B)$ for overlapping events.** The simple addition rule needs mutual exclusivity. Otherwise subtract the intersection.
2. **Forgetting the count in “exactly one”.** $\binom{n}{1}$ counts *which* one. “All” and “none” need no such factor.
3. **Binomial where the problem says “without replacement”.** That phrase means hypergeometric, and it changes the answer materially.
4. **Not reversing the inequality** when multiplying or dividing by a negative number.
5. $(AB)^t = A^t B^t$ and $(AB)^{-1} = A^{-1} B^{-1}$. Both reverse the order. Both are worth a whole question.
6. **Forgetting to transpose the cofactor matrix** when forming the adjoint. Check an *off-diagonal* entry of AA^{-1} , not just a diagonal one.
7. **Not computing D before applying Cramer’s rule.** If $D = 0$ the rule does not apply and the correct answer is a statement about the system.
8. **Stopping at $f' = 0$.** The second-derivative test is separately marked, and it is what distinguishes maximum from minimum from saddle.

B.2 Six habits worth the seconds they cost

Substitute your answer back. Nearly every solution in this document ends with a check line. It costs fifteen seconds and catches sign errors, which are the most common and the most expensive.

Tabulate before you compute. The probability questions in Unit 1 and the matrix word problems in Unit 5 become trivial once the data is in a two-way table, and nearly impossible without one.

Define your variable in writing. “Let n = the number of \$0.60 decreases.” Most marks lost in Unit 9 word problems are lost before any calculus starts.

Simplify algebraically first. $A + 2(B - A)$ is $2B - A$; a common factor in the numerator of a partial derivative often collapses the whole expression. Do the algebra before the arithmetic.

Sanity-check the units and the sign. A negative quantity, a probability above 1, an average cost below marginal cost — each is a signal to go back, and saying so in your script earns credit even if you cannot find the slip.

Show the method even when stuck. AIOU awards method marks. A correct formula with a wrong number scores; a right number with no working often does not.

B.3 If revision time is short

Units 4, 5, 6, 7 and 9 carry most of the marks and all of the long questions. Within those, the five highest-yield techniques are:

1. Cramer’s rule on a 3×3 system, including the $D = 0$ case.
2. Inverse of a 3×3 matrix by cofactors, with the adjoint transposed.
3. Product, quotient and chain rules — especially chain inside quotient.

4. Optimising profit, revenue or average cost, with the second-order test.
5. Finding a line's equation from two points, or from a point and a perpendicularity condition.

Work each of these from the relevant unit in this document with the solution covered. Five techniques, done cold and correctly, are worth more than nine units read passively.